

Swinburne University of Technology Sarawak Campus  
Faculty of Engineering, Computing and Science



MME 40001 Engineering Management 2

## **Assignment 1- Business Plan Project**

### **Final Report**

Group 2

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# Executive Summary

ThermaCorex Sdn Bhd aims to revolutionise personal beverage convenience in Malaysia through its smart thermal flask equipped with an advanced internal heating element. The business delivers high-quality, temperature-controlled drinkware designed for tech-savvy students, professionals, travellers, and health-conscious consumers who value safety, portability, and reliability. By addressing common challenges such as rapid heat loss, inconsistent beverage temperatures, and limited access to hot water, ThermaCorex provides a modern solution powered by intelligent thermal regulation technology.

The integration of smart temperature monitoring ensures optimal heating performance, enhances user experience, and supports energy-efficient operation, positioning ThermaCorex as a pioneer in Malaysia's smart consumer electronics segment. The comprehensive business plan presents strong financial strategies, scalable production frameworks, and a clear technology roadmap that supports long-term growth and product expansion. Although initial R&D and manufacturing costs are relatively high, financial projections indicate a shift toward profitability by 2027, driven by production optimisation, increased market acceptance, and strategic cost management. Key success factors include leveraging manufacturing partnerships, securing technology grants, and targeting premium lifestyle markets with value-driven product features.

This business plan addresses emerging consumer needs for convenience, sustainability, and smart living while aligning with national goals for advancing local tech innovation. ThermaCorex is poised to make a significant impact on Malaysia's consumer electronics landscape through its innovation-focused approach and well-structured execution strategy.

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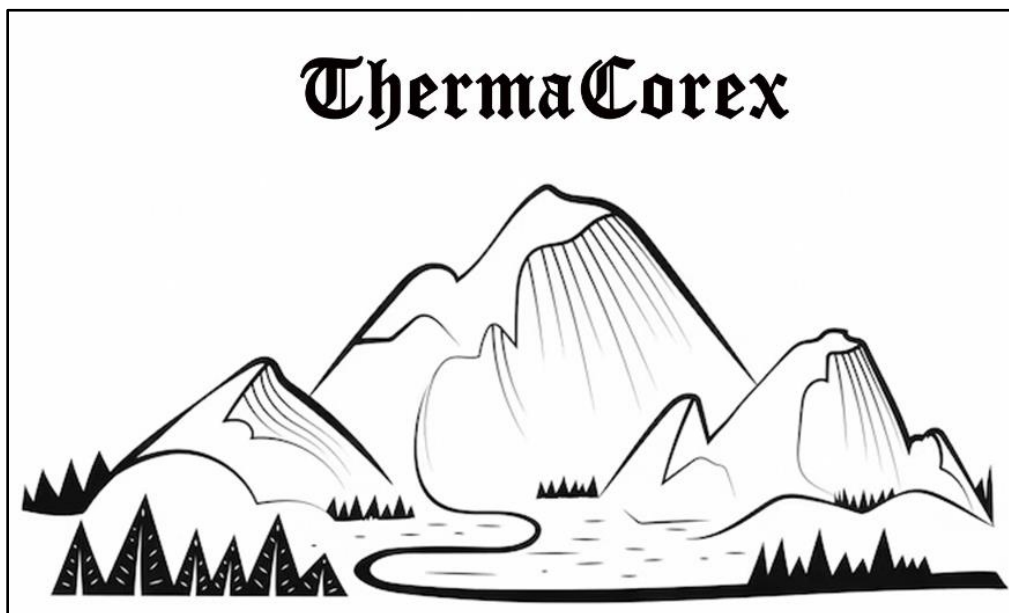
# 1.0 Company and Industry Background

## 1.1 Company Overview

ThermaCorex is a startup that focuses on producing smart vacuum flasks that support healthier and more sustainable hydration habits. The company develops products that combine reliable thermal performance with simple smart features such as temperature display, hydration reminders and sensor-based tracking. These features are designed for modern consumers who value convenience, technology and environmentally responsible alternatives to disposable plastic bottles.

ThermaCorex aims to deliver a better hydration experience through thoughtful design and sustainable practices. The company positions itself to meet the increasing demand for reusable smart drinkware that supports health, convenience and environmental awareness.

## 1.2 Company Name and Logo



*Figure 1 Company Logo which reflects slogan*

The company is named ThermaCorex to reflect both the functional capabilities of the smart vacuum flask and the core values the company aims to deliver to consumers. The term “Therma” signifies thermal insulation, temperature retention and overall heat management. These are the fundamental attributes of a high-quality vacuum flask. This reinforces the brand’s commitment to ensure beverages remain at their optimal temperature for extended period. The second component, “Corex,” is derived from the words core and complex, representing the integration of advanced smart technology at the heart of the product. It symbolizes the fusion of essential daily hydration with modern engineering, intuitive digital features and user-centric innovation.

Together, the name ThermaCorex conveys a brand identity centred on reliability, technological advancement and premium craftsmanship. It is distinctive, easy to pronounce and remember, giving it qualities that strengthen market presence and support long-term brand recognition. The name also aligns with the company's value proposition of offering smart, eco-friendly, and health-focused drinkware designed to enhance modern lifestyles.

The ThermaCorex logo features a stylised mountain range and flowing water. The mountain imagery symbolizes endurance, purity and resilience, attributes associated with nature and high-performance outdoor gear. This visual metaphor reflects the product's durability and ability to maintain beverage temperatures in any environment, whether at work, in transit or outdoors. The flowing stream illustrates hydration and continuous movement, aligning with the company's vision of encouraging healthier, on-the-go lifestyles. These natural elements incorporated into the logo reinforce the brand's commitment to sustainability and wellness, which are central themes in today's consumer market for reusable drinkware. Hence the slogan; "Hot or Cold, Anytime Anywhere".

The minimalist black-and-white design reinforces the brand's emphasis on clarity, precision and modern engineering. Overall, the combination of the logo elements visually communicate the company's dedication to quality, sustainability and high-performance technology, strengthening the overall brand narrative and supporting ThermaCorex's differentiation strategy

### 1.3 Business Model

ThermaCorex will operate under a hybrid manufacturer and direct-to-consumer (D2C) business model. Core manufacturing processes specifically the production of vacuum flask shells will be outsourced to reputable regional OEM partners, allowing the company to leverage established industrial capabilities while minimizing initial capital expenditure. Internally, ThermaCorex will focus on product differentiation through the design, innovation and integration of smart features, including IoT electronics, display modules and mobile app connectivity. Final assembly, quality testing and packaging will be conducted locally by our in-house to ensure precision, reliability and consistent product quality.

During the early operational years, the company will prioritize e-commerce channels as the primary sales platform to maintain low overheads, strengthen brand visibility and reach a wider consumer segment efficiently. As the business scales, ThermaCorex aims to expand into lifestyle retail outlets, corporate wellness programs and strategic partnerships with sports, healthcare and outdoor brands to broaden market reach.

The company's revenue will be generated primarily through sales of the smart vacuum flask, complemented by additional revenue streams such as replacement lids, battery modules, upgraded accessories, repair services and future premium app features. This multi-channel revenue structure supports long-term customer engagement and creates recurring value beyond the initial product purchase.

## 1.4 Target Customers

ThermaCorex targets a diverse yet strategically aligned consumer segment that values convenience, health, technology and sustainability. The primary market consists of tech-savvy millennials and gen Z consumers, who are early adopters of smart gadgets and appreciate products that integrate seamlessly into their digital and lifestyle ecosystems. These consumers prioritize personalization, app connectivity and real-time health insights, making the smart vacuum flask particularly appealing.

Another key segment includes fitness and wellness enthusiasts, such as gym-goers, hikers, cyclists and individuals pursuing active lifestyles. This group benefits from the flask's temperature retention, hydration reminders and smart-tracking features, which support daily routines and performance goals. Professionals and students also form an important demographic, as they often require durable, portable drinkware that maintains temperature throughout long workdays, study sessions or commutes. For these users, the product offers a balance of practicality, style and digital convenience.

In addition, ThermaCorex appeals strongly to eco-conscious consumers who prefer reusable, long-lasting alternatives to disposable plastic bottles. This segment is motivated by sustainability, environmental responsibility and the desire to reduce their ecological footprint, values that align closely with the company's mission and branding.

The secondary market includes corporate organizations, institutions and wellness-focused enterprises that may integrate ThermaCorex products into employee engagement, health promotion or sustainability initiatives. Companies looking to support workplace hydration habits or reduce single-use plastics may adopt the smart flask as part of corporate wellness programs, employee welcome kits or branded merchandise.

Overall, ThermaCorex's target market represents consumers who seek a combination of health, technology, practicality and sustainability, making the product well-positioned to capture demand within both individual and organizational customer segments.

## 1.5 Business Location

We selected Bayan Lepas, Penang as the location for ThermaCorex because it provides the most strategic and supportive environment for our business operations. The Bayan Lepas Free Industrial Zone is Malaysia's most established high-technology manufacturing hub and offers an exceptional level of industrial maturity, supply-chain integration and operational reliability. Although the rental cost in this area is higher than in developing zones, the advantages provided by its advanced ecosystem outweigh the additional expense. The district has a long history of hosting leading multinational companies such as Intel, HP, Broadcom and AMD. Their presence has created a highly skilled talent pool, strong engineering support services and a dense network of suppliers.

These conditions are ideal for a company that focuses on smart consumer products and the integration of electronics within a manufactured device.

Bayan Lepas is also distinguished by its excellent logistical connectivity. Its location adjacent to the Penang International Airport enables fast export processing and efficient inbound delivery of components. Freight forwarders, warehousing facilities and logistics providers are concentrated within the area, which supports reliable distribution and reduces transportation delays. This accessibility is particularly valuable for a business that plans to serve a national and potentially international market.

A comparison with Kuala Lumpur industrial zones such as Petaling Jaya, Kepong and Puchong further strengthens the decision. These locations offer modern infrastructure and access to a large urban market, but they experience significantly higher rental and labour costs. Traffic congestion is also a major operational barrier in the Klang Valley and contributes to delays in transportation and distribution. Kuala Lumpur's industrial ecosystem is more aligned to retail and commercial services rather than precision manufacturing and export-oriented production.

Penang presents a more suitable environment for ThermaCorex because it is recognized internationally as Malaysia's centre for electronics, precision engineering and high-technology manufacturing. Based on this, Bayan Lepas supports the long-term competitiveness of the company through its strong technical workforce, established supplier base, global connectivity and stable operating environment. These characteristics create an ideal foundation for the assembly, testing and distribution of our smart vacuum flask and position the business for sustainable expansion in the future.

## 1.6 Type of Industry

ThermaCorex operates within the secondary industry, specifically in the manufacturing sector, where it focuses on the development of smart vacuum flasks positioned within the broader consumer electronics and sustainable lifestyle market. The company's core function is to design and integrate advanced smart features such as temperature tracking, Bluetooth connectivity and hydration reminders into high-quality vacuum flasks. Rather than producing components from raw materials, ThermaCorex collaborates with established OEM manufacturers to fabricate the stainless-steel flask shells. This approach ensures cost efficiency, consistent quality and access to technologically advanced production capabilities.

By combining in-house innovation with outsourced manufacturing, the company is able to concentrate on product differentiation, user experience and sustainable design. This operational model also allows ThermaCorex to leverage existing global supply chains, reduce capital-intensive investment and accelerate time-to-market. Through this differentiation strategy, the company delivers eco-friendly and technologically enhanced products that align with contemporary wellness, environmental and lifestyle trends, positioning ThermaCorex competitively within the growing smart drinkware industry.

## 1.7 Business Entry Strategy

To start our own business to launch this unique smart vacuum flask gives us critical advantages in control differentiation, and long-term value. This is due to the customization on not just the appearance, shape, materials, branding and the customization on the assembling of the LED + heating components locally (including developing the GUI), thus this strategy ensures that we can maintain high quality, unique features protection, and also build a strong brand identity. Furthermore, full ownership allows us to innovate continuously, respond quickly to customer feedback, and ensure that every part of the product and user experience aligns with our vision. While starting from scratch requires higher up-front investment in R&D, assembly, certification, marketing, and infrastructure, research shows that these strategic branding resources significantly improve SME competitiveness in Malaysia. Thus, by anchoring our business model in uniqueness, innovation, and branding consistency, we can create a strong, differentiated market presence and capture long-term growth potential.

## 2.0 Mission and Objective of the Company

The mission of ThermaCorex company is to promote and encourage daily hydration by combining smart technology with sustainable design, making it easier for people to maintain a healthy lifestyle while reducing environmental impact. The company also aims to tackle common issues within the vacuum flask market such as heavy weight and bulky size.

The objectives of ThermaCorex company are as follows:

1. Launch ThermaCorex in the Malaysian market within 18 months and achieve positive consumer adoption.
2. Capture 5% of Malaysia's reusable bottle market within three years.
3. Continuously innovate by introducing new features such as AI-driven hydration tracking, enhanced customization options and advanced build materials within a structured three-year development roadmap:
  - Release incremental feature updates annually, beginning with sensor improvements and firmware upgrades in the first year
  - Release AI-supported hydration analytics and app integration in Year 2
  - Targets the implementation of higher-grade sustainable materials and modular design options to further strengthen product differentiation and maintain a competitive advantage in the smart drinkware market.
4. Promote environmental sustainability by achieving a 15–20% conversion rate among target customers from disposable plastic bottles to reusable smart drinkware within the first three years of operation.

## 3.0 Legal Requirements

To legally operate in Bayan Lepas, Penang, ThermaCorex must obtain several mandatory licences and permits issued at the federal, state and municipal levels. These approvals ensure that the company's assembly, warehousing and commercial activities comply with Malaysian regulatory standards.

### 3.1 Business Licence

#### **Business Premise Licence**

As ThermaCorex will operate from an industrial unit in Bayan Lepas, the company must secure a business premise licence from the Penang Island City Council (MBPP). This licence authorises the use of the selected premise for light industrial activities such as assembly, storage and administration. It verifies compliance with zoning regulations, building safety requirements and public health standards (MBPP 2025).

#### **Trade Licence**

The company must also obtain a trade licence, which permits ThermaCorex to conduct commercial activities including the distribution and sale of consumer electrical products. This licence is required for all trading operations within Penang and is renewed annually.

#### **Signboard Licence**

If ThermaCorex displays external or internal signage, a signboard licence must be obtained from MBPP. Approval is needed for the placement, design and content of the company's signboard to ensure compliance with local advertising regulations.

#### **SSM Company Incorporation (Federal Requirement)**

Before applying for municipal licences, ThermaCorex must be incorporated as a Sendirian Berhad (Sdn. Bhd.) under the Companies Commission of Malaysia (SSM) through the MyCoID 2016 system (Ramsay et al. 2006). This step legally establishes the company and enables it to engage in manufacturing, trading and hiring activities.

#### **Manufacturing Licence**

Under the Industrial Coordination Act 1975, a manufacturing licence is only required for manufacturers with at least:

- 75 full-time employees
- Shareholder funds  $\geq$  RM2.5 million.

As ThermaCorex operates on a light-assembly model with fewer than 75 employees, the business is exempted from a manufacturing licence but must apply for one if future expansion exceeds these thresholds (MIDA 2025).

### 3.2 SIRIM Certification

Since the smart vacuum flask contains a heating element, LED display and rechargeable battery, it is categorized as an electrical and electronic (E&E) consumer product. As such, the product must undergo SIRIM QAS Product Certification and receive approval from the Energy Commission (Suruhanjaya Tenaga) before importation or sale (Suruhanjaya Tenaga 2018; SIRIM QAS 2025). This process involves submission of technical documents, safety testing, and labelling each unit with the official SIRIM hologram. SIRIM certification ensures safety compliance, prevents customs delays, and enhances consumer confidence.

As our smart vacuum flask incorporates an LED timer frequency display, rechargeable battery and heating element, it is classified as an electrical and electronic (E&E) consumer product in Malaysia. To ensure compliance with national standards, the business is required to obtain SIRIM certification prior to importation and sale. This certification verifies that the product adheres to Malaysian safety and technical requirements. (Suruhanjaya Tenaga 2018)

The certification process involves an application to SIRIM QAS International, submission of technical specifications, and possible product testing to confirm compliance with MS/IEC standards. Once approved, the smart vacuum flask must carry the official SIRIM hologram label before being released to the market.

For the business, securing SIRIM certification is not only a legal requirement but also a strategic advantage, as the SIRIM mark is widely recognized by Malaysian consumers as a symbol of product safety and quality. Certification ensures smooth customs clearance, prevents regulatory issues and enhances the brand's reputation. By integrating SIRIM certification into our operations, we can confidently introduce our smart vacuum flask into the Malaysian market while assuring customers of its safety and reliability.

### 3.3 Trademarks

To protect our brand identity and prevent imitation, the company will register a trademark with the Intellectual Property Corporation of Malaysia (MyIPO). This includes the brand name, logo, and any unique product design elements such as the LED display pattern or slogan.

Registering a trademark provides the company with:

- Exclusive rights to use the brand in Malaysia, preventing competitors from copying or misusing it.
- Legal protection against counterfeit or “copycat” smart flasks entering the market.
- Enhanced brand value when approaching distributors, corporate buyers, and e-commerce

platforms.

- Long-term asset creation, as trademarks can be renewed every 10 years and form part of the company's intellectual property portfolio.

By securing trademark protection early, the business safeguards its brand image and builds stronger consumer trust, ensuring that customers can clearly identify our smart vacuum flask in a competitive market.

### 3.4 Business Registration Process

The establishment of ThermaCorex begins with formal registration under the Companies Commission of Malaysia (SSM), the statutory body responsible for regulating corporate entities under the Companies Act 2016 (Ramsay et al. 2006). As the company will operate as a private limited entity, Sendirian Berhad (Sdn. Bhd.), the incorporation procedure is completed through SSM's online MyCoID 2016 system.

Under the Companies Act 2016, the incorporation of a Sdn. Bhd. is performed using the Superform (Section 14: Application for Incorporation), which consolidates the previous manual forms such as Form 13A, Form 48A and the Memorandum & Articles of Association (SSM n.d.). The Superform allows the company to submit its proposed name, business nature, registered address, shareholder and director details and if desired an optional company constitution (OCBC Business Banking MY n.d.). A Sdn. Bhd. requires a minimum of one director who is ordinarily resident in Malaysia and at least one shareholder, both of whom may be the same individual.

A digital name search is conducted within the Superform system to confirm that the proposed name complies with SSM naming rules, ensuring that it is not misleading, vulgar, overly generic or identical to an existing entity. Once approved, the incorporation fee of RM1,000 is paid online, after which SSM issues the Notice of Registration, officially recognising ThermaCorex Sdn. Bhd. as a legally incorporated entity in Malaysia.

Following incorporation, the company must appoint a qualified company secretary within 30 days, as required under the Companies Act 2016. ThermaCorex is then obligated to maintain statutory compliance through annual return submissions, corporate tax filings, employee social security contributions (EPF, SOCSO) and updates on company particulars (Perkeso 2023).

Because the business will operate from a light industrial unit in Bayan Lepas, it must also obtain the required municipal operating licences. These include the Business Premise Licence and Trade Licence, both issued by the Penang Island City Council (Majlis Bandaraya Pulau Pinang). These licences certify that the chosen industrial facility is approved for assembly, warehousing and commercial activities. This also ensures that operations comply with local zoning, safety, building and public health regulations (MBPP 2025). Approval from MBPP is mandatory before ThermaCorex may commence manufacturing, assembly or distribution activities within Penang Island.

## 4.0 Product Characteristics

### 4.1 Product Uniqueness and Distinctiveness

The smart vacuum flask developed by our company is distinctive because it combines both design and technological enhancements into a single product. Unlike conventional flasks that only provide insulation, our product features a customized shape, materials, and branding that give it a unique identity. In addition, it incorporates an LED screen with a custom graphical user interface (GUI) and a heating element, additionally it also contains a rechargeable battery to maintain the user's desired temperature, both of which are integrated and assembled locally by our team. This integration not only improves functionality but also differentiates our smart vacuum flask in the Malaysian market, offering users the ability to check beverage temperature instantly and reheat drinks when required.

Given these distinctive features, intellectual property (IP) protection is essential to safeguard the business. The industrial design of the flask (its shape, form, and external look) can be registered with the Intellectual Property Corporation of Malaysia (MyIPO) to prevent imitation of its physical appearance. The brand name and logo should also be protected under trademark registration with MyIPO, ensuring exclusive rights to our brand identity within Malaysia. Additionally, the custom GUI design for the LED screen may qualify for copyright protection, also under MyIPO, securing the originality of the user interface created by our team.

Although the LED and heating technology are imported, our role in customizing, assembling, and branding the product creates a unique value proposition. By applying for the appropriate IP rights through MyIPO, the company not only ensures legal protection against local competitors but also enhances its reputation for innovation and quality. This strategic approach strengthens the business in the Malaysian market by making the smart vacuum flask both distinctive and defensible, while also positioning the brand for future growth opportunities.

## 5.0 Marketing Analysis and Strategy

The proposed product is a Smart Vacuum Flask that integrates the functions of a traditional insulated flask with modern electronic features. Unlike conventional vacuum flasks which function purely as passive insulated containers designed to maintain temperature through double-walled vacuum insulation, the ThermaCorex smart vacuum flask incorporates active thermal and digital features that significantly enhance functionality. In addition to standard insulation, the smart flask integrates a miniature heating coil capable of boiling or reheating water, enabling users to achieve on-demand temperature control without an external heat source. It is powered by a USB-C rechargeable lithium-ion battery, providing true portability compared to traditional flasks that lack

any electrical components. Furthermore, the smart flask includes an LED touchscreen display that reports real-time internal temperature, issues hydration reminders and supports light smart-device interactions, features entirely absent in conventional stainless-steel or plastic insulated bottles (Research Nester, 2024; Maximize Market Research, 2024). This unique combination creates a hybrid device that goes beyond simple storage, offering both convenience and safety for on-the-go users. Comparable products in the market include standard vacuum flasks such as ThermoFlask, Lock&Lock, and Hydro Flask, as well as portable travel kettles from brands like Xiaomi and Philips (6WResearch 2023). Our product strategy is focused on differentiation, positioning the smart vacuum flask as a premium yet affordable lifestyle gadget that combines utility, technology, and modern design (MIDA 2024).

The product exists within the consumer electronics and lifestyle drinkware industry, which is highly competitive and dominated by e-commerce sales (MIDA 2024; Datacube Research 2024). This market structure provides both opportunities and challenges, as online platforms reduce entry barriers but also increase exposure to cheaper alternatives (CapillaryTech 2024). In its first 3 years, the smart flask aims to capture an estimated 5% of the market (about 500,000 units). With effective branding and partnerships, market share is projected to reach approximately 0.5% of the targeted market (about 50,000 units) within its first year since it is considered a start-up.

Marketing channels will be focused on an e-commerce-first strategy, leveraging platforms such as Shopee, Lazada, and TikTok Shop to maximize reach at low distribution cost (PaymentsCMI 2024; Incorp Malaysia 2024). An official website will also be developed for direct sales and customer support, while partnerships with corporate organizations will provide opportunities for bulk orders and brand visibility. In the long term, expansion into lifestyle retail outlets such as Parkson, AEON & Harvey Norman will increase accessibility to offline customers (MIDA 2024).

The promotion strategy is designed to be segment specific. For students and Gen Z consumers, promotions will emphasize TikTok or Instagram campaigns, influencer collaborations, and student ambassador programs (CapillaryTech 2024). Professionals and corporate clients will be reached through LinkedIn campaigns, bundled corporate gift offerings, and wellness program tie-ins. Fitness and wellness enthusiasts will be targeted via Instagram or Facebook ads, sponsorship of fitness events, and partnerships with gym influencers. Parents and frequent travellers will be reached through Facebook marketing, product reviews from parenting bloggers, and messaging that highlights safety and portability. This segmented promotion strategy ensures that the brand resonates with diverse groups while maximizing awareness (6WResearch 2023).

Finally, the proposed pricing strategy positions the smart vacuum flask in the RM120–150 range. This pricing balances affordability for Malaysian consumers with the higher production costs associated with advanced features and certification (Research Nester 2024). The range is competitive compared to portable kettles (RM100–200) and premium insulated flasks (RM150–300) (Maximize Market Research 2024). To stimulate early adoption, the product can be

introduced with a launch promotion at RM99 for early bird buyers. Corporate bulk orders will be offered at discounted rates, allowing the product to penetrate multiple markets simultaneously. This balanced strategy ensures that the flask is positioned as a premium yet accessible product in Malaysia's consumer market.

## 5.1 Target Market

The target market for this product consists primarily of tech-savvy Millennials and Gen Z consumers aged 18 to 35 who frequently purchase lifestyle gadgets and smart accessories through online platforms. This demographic is highly familiar with digital technology and places strong value on convenience, personalisation and modern product design. The segment represents a major portion of the Malaysian online consumer base and continues to expand as younger individuals adopt e-commerce as their main purchasing channel (Population Trends Asia Pacific 2024).

Corporate organisations form another important market. Many companies invest in workplace wellness programmes and employee engagement initiatives and these organisations may purchase the smart flask in bulk as a corporate gift or as part of wellness-driven incentive packages. Corporate procurement is an attractive segment because it provides higher volume sales and enhances brand visibility among working professionals.

Outdoor and travel enthusiasts represent a growing opportunity. These consumers frequently participate in activities such as hiking, camping and sports. They value portable and durable hydration solutions that support an active lifestyle. The rise in outdoor recreation in Malaysia has strengthened demand for products that combine functionality with reliability (6WResearch 2023). The smart flask serves this need by offering temperature retention and digital features that support hydration during extended outdoor use.

Professionals such as office workers, parents and frequent travellers form another niche segment. These users require convenient and portable access to hot water throughout the day. They also prioritise safety, reliability and ease of use. Their daily routines often involve commuting between locations which increases the need for products that offer consistent performance.

Demographically, the core target audience consists of consumers aged 18 to 35 living in urban and suburban regions. They generally possess moderate to high disposable income and purchase regularly from e-commerce platforms. They are typically trend conscious and prefer products that combine practicality with modern design. In Malaysia, this demographic represents approximately 10 million individuals which accounts for 27.8 percent of the population (Department of Statistics Malaysia 2024; IndexMundi 2024).

The target market aligns well with current trends in sustainable consumption, digital adoption and active lifestyle behaviour. Consumers in this age group are increasingly aware of environmental issues and prefer reusable drinkware solutions over single-use plastics. They also demonstrate a high willingness to try new smart products that provide added convenience and functionality. This

combination of behavioural and demographic characteristics creates strong potential for adoption and long-term brand loyalty.

## 5.2 Target Market Size

Based on the latest Malaysian demographic data, the total population stands at 34.2 million people (OpenDOSM 2025). In a survey titled “Analysis of consumer preferences and attitudes towards sustainable consumption of plastic beverage packaging in Malaysia”, 74.4% of Malaysian respondents preferred reusable bottles over single-use plastic drink containers, reflecting strong alignment with sustainability trends (Bakar 2024). From the survey, 42.3% of Malaysians fall within the Gen Z and Millennial age groups, which represent the primary adopters of lifestyle electronics and smart consumer products. E-commerce readiness further refines the target market where national surveys indicate that 45–55% of urban working adults regularly purchase personal electronics online and a reasonable assumption of 50% online purchasing behaviour is applied for this analysis (MCMC 2023).

Table 1 First-Year New Tech Adoption Rate from Different Sources

| Source              | First-Year New Tech Adoption Rate |
|---------------------|-----------------------------------|
| DOSM                | 0.4–1.2%                          |
| Statista Smart Home | 0.1–0.9%                          |
| IDC Tracker         | 0.3–0.9%                          |

To estimate the potential first-year adoption of a new smart consumer device, the first-year new tech adoption rate from Department of Statistics Malaysia (DOSM), Statista and IDC collectively show an adoption band of 0.4% to 0.9%, with this business plan applying a conservative median adoption rate of 0.7% for the ThermaCorex smart vacuum flask. Combining these factors, the estimated target market size for first year is calculated as:

$$\begin{aligned}
 \text{Estimated Target Market Size} &= 74.4\% \times 42.3\% \times 50\% \times 0.7\% \times 34.2M \\
 &= 37671.03 \\
 &\approx 37500 \text{ consumers}
 \end{aligned}$$

## 5.3 Marketing Mix (4Ps)

### Product

The smart vacuum flask is designed to go beyond traditional flasks by integrating smart features such as an LED temperature display, hydration reminder timer, and heating element. Constructed with high-grade stainless steel, it ensures durability, safety, and insulation. The modern and minimalist design aligns with consumer preferences for stylish and functional everyday items.

The product delivers value-added features that appeal to health-conscious individuals who want to monitor water temperature and stay consistent with their hydration habits. With its multipurpose use in work, gym, travel, and outdoor activities, the flask positions itself as both a practical utility and a lifestyle accessory.

### **Price**

The price range is set at RM150, which reflects a competitive yet affordable positioning within the smart product market. This range makes the product accessible to middle-income consumers while maintaining sufficient margins for business sustainability.

The pricing strategy also incorporates promotional discounts such as early bird launch offers, bundled sales (e.g., flask + carrying case), and limited-time campaigns to attract first adopters. Over time, the company can introduce premium variants or accessories to cater to different customer segments.

### **Place (Distribution)**

Distribution will be focused on e-commerce platforms such as Shopee, Lazada, and TikTok Shop, which are widely used by the target demographic. In addition, a dedicated company website will provide direct sales opportunities and serve as a central hub for branding, product information, and customer support.

As the brand gains traction, expansion into physical retail channels such as lifestyle stores, gyms, and outdoor equipment shops will further strengthen market presence. This omnichannel strategy ensures broad accessibility and enhances brand credibility.

### **Promotion**

Promotional activities will emphasize building strong brand awareness and creating lifestyle appeal. The strategy includes:

- **Social Media Marketing:** Active campaigns on Instagram, TikTok, and Facebook showcasing the product's smart features and portability through engaging content.
- **Influencer Collaborations:** Partnering with micro-influencers in fitness, travel, and tech to create authentic product reviews and lifestyle demonstrations.
- **Content Marketing:** Hydration challenge videos, tutorials, and customer testimonials to highlight the importance of hydration and the benefits of the product.
- **Launch Campaign:** "Early Bird Specials" offering limited discounts and giveaways to stimulate buzz and encourage trial.
- **Customer Loyalty Programs:** Referral rewards, user-generated content contests, and engagement campaigns to strengthen community ties and promote word-of-mouth marketing.

## 5.4 Positioning and Branding

The smart vacuum flask will be positioned as “a lifestyle companion, not just a bottle.” Unlike conventional flasks that merely store beverages, this product integrates technology, health benefits, and style into a single solution.

Branding efforts will emphasize innovation, reliability, and modern living, appealing to consumers who see hydration as part of a healthy, efficient, and active lifestyle. The SIRIM certification and compliance with safety standards will further reinforce trust and credibility among Malaysian consumers.

## 5.5 Sales Forecast and Growth Strategy

In the initial phase, sales are expected to be driven primarily through early adopters and online campaigns, leveraging the popularity of e-commerce platforms. Within the first year, the business anticipates steady growth as consumer awareness increases through social media engagement and influencer marketing.

Subsequent growth will come from:

1. Expanding distribution channels to include physical retail outlets.
2. Launching premium versions or accessories to capture higher-value customers.
3. Leveraging word-of-mouth referrals and positive reviews to sustain long-term demand.

The marketing strategy aims to build a sustainable and recognizable brand, ensuring that the smart vacuum flask is perceived as an essential companion for modern, on-the-go lifestyles.

## 6.0 Competitive Analysis

### 6.1 SWOT Analysis

The SWOT analysis helps to highlight both the internal capabilities and external conditions that shape the competitiveness of the ThermaCorex smart vacuum flask. The product carries several important strengths. It introduces smart features such as temperature display, hydration reminders and a rechargeable power system that distinguishes it from traditional insulated bottles (Research Nester 2024). The reusable nature of the flask matches the growing shift toward sustainability. The product can fit easily into many situations which include office use, gym sessions and outdoor activities. The strong presence of e-commerce in Malaysia provides an accessible platform for reaching younger consumers who are highly active online (CapillaryTech 2024).

The product also has weaknesses that the company must address. The inclusion of electronic components raises production costs, which increases the financial burden for a new business.

Certification requirements such as SIRIM testing extend the time needed to bring the product to market (SIRIM 2024). The company depends on OEM partners for the production of the stainless steel shell which reduces its level of control over some parts of the manufacturing process.

There are several opportunities for ThermaCorex to grow. Health awareness is rising among Malaysian consumers which increases demand for hydration related products. Environmental concerns have encouraged more people to choose reusable solutions (PaymentsCMI 2024). The increase in local travel and outdoor recreation strengthens interest in portable and durable drinkware (6WResearch 2023). Growth in e-commerce platforms also creates a favourable environment for nationwide market reach at relatively low cost (Datacube Research 2024).

The company must be prepared for several threats. Established brands in the insulated bottle and kettle market have strong customer loyalty and large distribution networks. Many low priced alternatives are available on Shopee and Lazada which may influence consumers who are sensitive to price (Incorp Malaysia 2024). External challenges such as regulatory delays, exchange rate fluctuations and supply chain disruptions can affect the import of electronic components (TecEx 2024).

## 6.2 Porter's Five Forces Analysis

Porter's Five Forces analysis provides a clearer view of the competitive environment for the smart vacuum flask. Industry rivalry is strong because the market includes well known brands that produce insulated drinkware and portable kettles. These brands already have strong distribution channels which intensify competition (6WResearch 2023). The threat of substitutes is high because consumers may rely on standard flasks or compact kettles that serve similar purposes at lower cost (Tracedata Research 2024).

The threat of new entrants is medium. Integrating electronics requires higher development costs although many start-ups continue to enter the consumer electronics space due to easier access to OEM manufacturing and online marketing channels. Larger brands may also replicate the concept once it proves successful (MIDA 2024). Supplier power is moderate. Batteries, sensors and electronic modules can be sourced from many suppliers yet imported components remain vulnerable to shipping and regulatory challenges (TecEx 2024).

Buyer power is high in Malaysia. Consumers frequently compare prices and read reviews before making purchases. Online platforms make it easy for them to switch between brands which places pressure on new entrants to deliver strong value and reliability (PaymentsCMI 2024). This analysis highlights the need for ThermaCorex to differentiate through technology, quality assurance and consistent user experience.

## 6.3 Risk Management

Strong risk management practices are essential for the sustainability of ThermaCorex. The business faces operational risks related to supplier dependency. The company relies on OEM

partners for stainless steel shells and imported electronic components. Shipping delays or production interruptions may slow down assembly and reduce product availability. A practical approach is to diversify suppliers and maintain safety stock for key components (TecEx 2024).

Regulatory risks are also significant since the smart flask requires certification through SIRIM before it can enter the Malaysian market. Delays in approval may affect launch schedules. Early submission and consultation with compliance specialists can reduce this risk (SIRIM 2024). Financial risks arise from exchange rate fluctuations which influence the cost of imported materials. Approaches such as negotiated long term pricing and maintaining financial reserves can provide stability.

Market risks relate to customer perception and competition. Lower priced alternatives on online platforms create pressure for competitive pricing and clear product differentiation (Incorp Malaysia 2024). Educating customers about the value of smart features and offering warranty programmes can reduce hesitation among buyers. Technological risks include potential malfunction of electronic components which may lead to negative customer experiences. Rigorous quality testing before shipment can lower the likelihood of defects.

Reputational risk is also important because customer reviews on social media influence purchasing decisions. A responsive customer service system that resolves issues quickly can protect the brand. Strategic risks are tied to shifts in consumer trends and expectations. Continuous product improvement and alignment with sustainability and wellness trends are essential for long term relevance (CapillaryTech 2024).

## 7.0 Operations Plan and Management

### 7.1 Quality Function Deployment (House of Quality)

The House of Quality (HoQ) is a central component of Quality Function Deployment (QFD), serving as a structured method for translating customer needs into measurable engineering specifications. It visually resembles a house, where the foundation represents customer requirements, the walls correspond to technical attributes, and the “roof” forms the correlation matrix showing interrelations among these attributes (ASQ 2024). In this project, the HoQ was constructed to guide the design of a next-generation smart thermal flask that prioritises insulation performance, reliability, safety, sustainability, and smart functionality. Figure 1 illustrates the developed HoQ model, which aligns customer expectations with specific design and production targets.

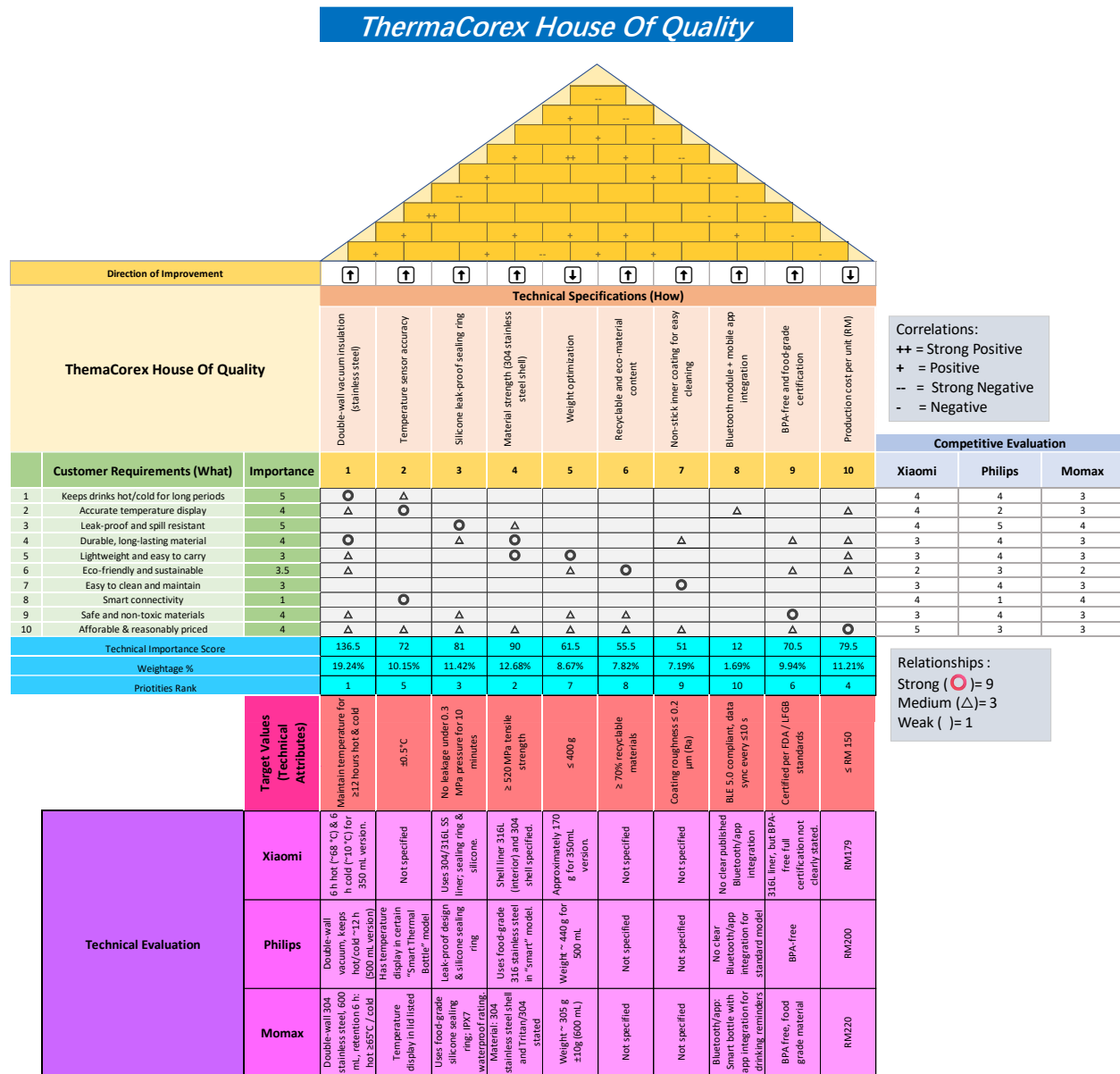


Figure 2 House of Quality

Due to limited access to proprietary industrial data, the team based its technical targets and benchmarking analyses on publicly available specifications and estimated performance data from leading competitors such as Xiaomi, Philips and Momax. Based on the completed HoQ, the most significant technical priorities identified for the product are thermal insulation efficiency, temperature sensor accuracy and leak-proof reliability. These are followed by other important factors such as lightweight construction, BPA-free certification, sustainability through recyclable materials and Bluetooth connectivity for data monitoring. Together, these attributes form the foundation for a practical and consumer-driven product design approach.

## **Identifying Customer Needs**

The first stage of the HoQ process involves identifying and prioritising customer needs, often referred to as the “WHATs.” In the context of the smart flask market, these needs were determined through observation of current lifestyle trends, user reviews, and competitive benchmarking. Consumers increasingly seek products that combine convenience, health safety and environmental sustainability. Key customer requirements include the ability to maintain beverage temperature for extended periods, a leak- proof and durable design for everyday portability, a lightweight body for ease of handling and include modern smart features such as temperature display or optionally Bluetooth connectivity. Furthermore, users demand products made from eco-friendly, food-safe materials that are easy to clean and maintain, all while remaining reasonably priced. These customer insights establish the baseline expectations that drive the technical design of the proposed smart flask.

## **Product Attributes (Technical Requirements)**

The second stage translates customer needs into technical attributes, the “HOWs” that engineers can measure, evaluate and optimise in the design. For the smart flask, ten key attributes were identified to align with consumer priorities. These include double-wall vacuum insulation to maintain temperature for at least twelve hours, a high-accuracy temperature sensor with a tolerance of  $\pm 0.5$  °C and a silicone sealing ring tested for leak resistance under 0.3 MPa of pressure. Material strength is ensured using 304 stainless steels with a tensile strength exceeding 520 MPa, while overall weight is targeted to remain below 400 g for improved portability. Sustainability goals specify the inclusion of at least 70 % recyclable material, complemented by a non-stick inner coating for easy cleaning and a smoothness level below 0.2  $\mu\text{m Ra}$ .

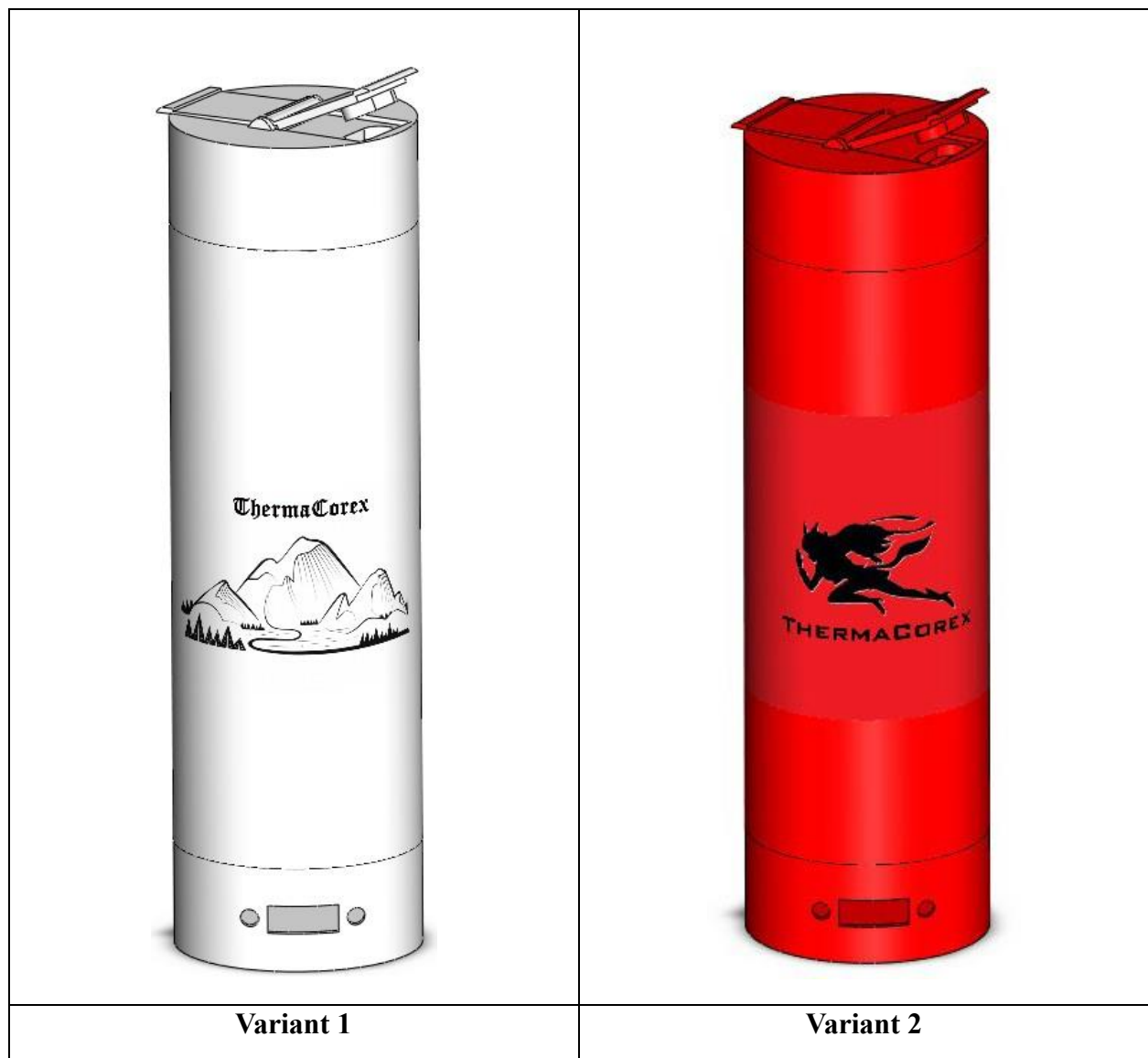
Additional smart features which we plan to implement into future versions include a Bluetooth 5.0 module with fast data synchronisation, ensuring seamless connectivity with a mobile application for real- time monitoring. All materials are expected to meet FDA and LFGB food-grade certifications to guarantee consumer safety. The production cost target is set around RM 120 per unit, balancing functionality with manufacturing feasibility. These attributes collectively serve as quantifiable goals to guide material selection, component integration, and production planning throughout the design process.

## **Benchmarking Against Competitors**

To position the proposed smart flask competitively, a benchmarking analysis was conducted against three leading brands which are Xiaomi, Philips and Momax. Xiaomi’s Smart Insulated Cup offers reliable double-wall vacuum insulation, an OLED temperature display and a minimalist design priced at RM179. Philips’ GoSmart Flask focuses on build quality and safety, featuring SUS304 stainless steel construction, a detachable lid, LED temperature display and BPA-free certification, typically priced between RM 170 and RM 200. Momax’s Smart Flask, in contrast, targets the premium market segment with features such as Bluetooth app integration, a

rechargeable lid sensor and IP66 water resistance, retailing for approximately RM 220 to RM 250. Through this comparison, it was found that Xiaomi's flask performs strongly in affordability, Philips excels in material safety and durability while Momax leads in smart functionality and connectivity. However, none of these brands fully integrate all three pillars; affordability, sustainability and smart functionality into a single product. The proposed design for this project aims to bridge that gap by combining Momax's advanced digital features, Philips' emphasis on quality and certification, and Xiaomi's cost efficiency. This strategic positioning enables the product to target both mainstream consumers and tech-conscious users who value long-term durability and environmental responsibility.

### **Consolidating the Product Design**



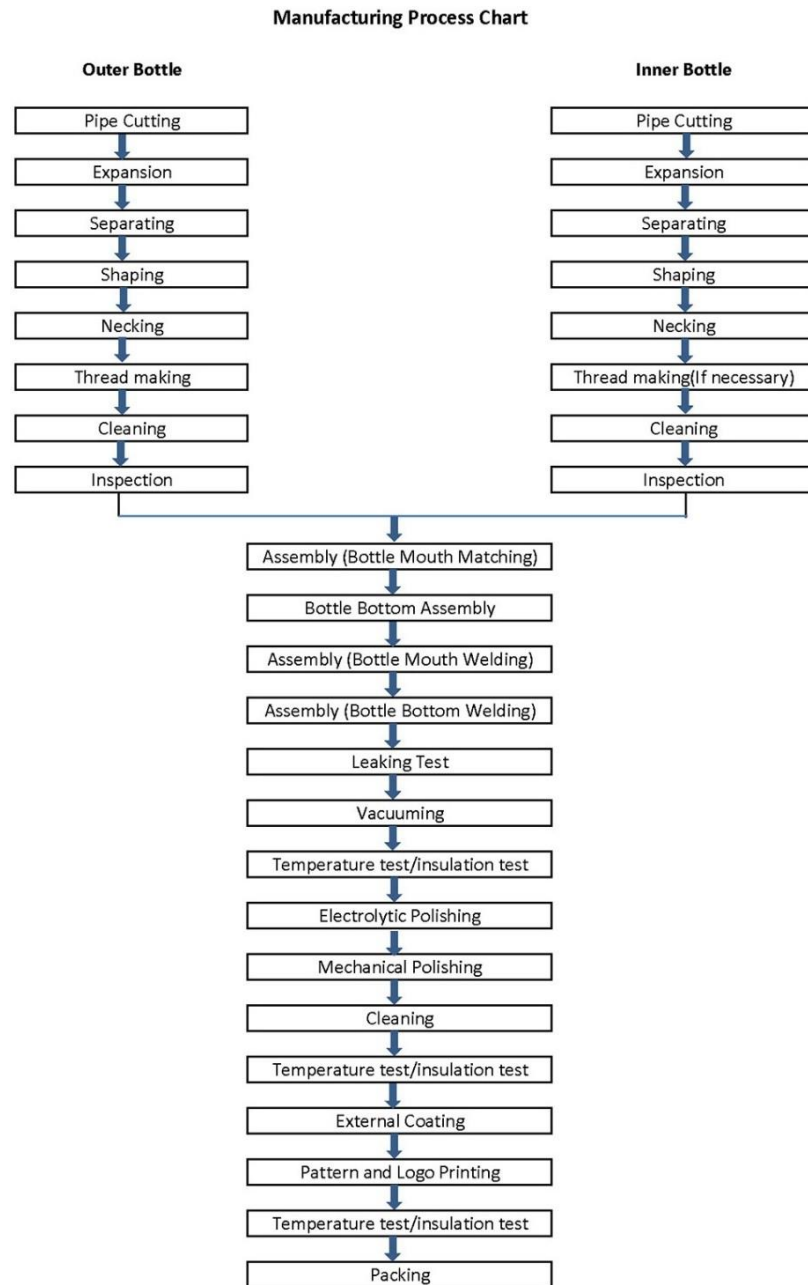
*Figure 3 ThermaCorex Product Design*

The final stage in the HoQ process is the consolidation of findings into a coherent product concept that reflects both customer desires and feasible technical execution. The design prioritises enhanced insulation performance through refined double-wall vacuum technology and precise temperature sensing using thermistor-based digital modules. A leak-proof silicone seal ensures usability under pressure, while lightweight and sustainable materials maintain portability without compromising strength. The inclusion of Bluetooth 5.0 connectivity and potential app integration adds value by enabling real-time temperature tracking and hydration reminders which are appealing features for health-conscious and tech-savvy consumers.

Although Bluetooth functionality has not yet been implemented in the current prototype, it is identified as a future enhancement in subsequent iterations, aligning with long-term product development goals. By consolidating these elements, the smart flask embodies a well-balanced design that integrates performance, safety, sustainability, and affordability. This systematic process demonstrates how the HoQ methodology effectively transforms abstract customer expectations into targeted concrete technical solutions that can be achieved through modern engineering and manufacturing.

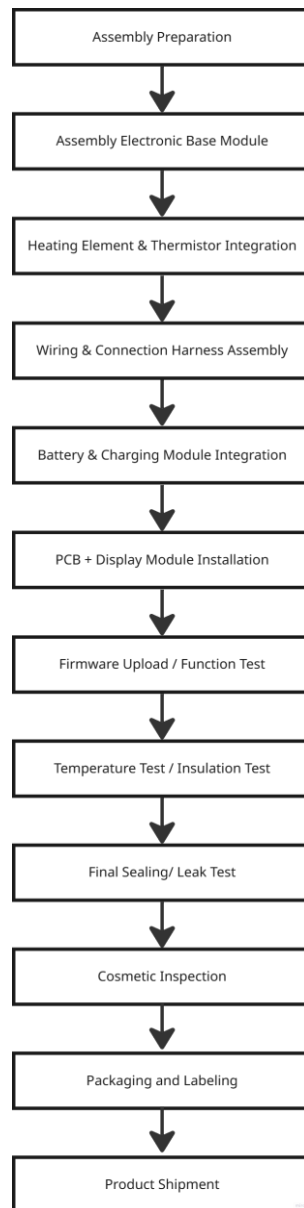
## 7.2 Production Process & Flow Chart

The existing production of smart beverage containers and electronic smart vacuum flasks generally follows a semi-automated assembly line process that integrates both mechanical and electronic components. Manufacturers of products such as smart kettles, temperature-controlled mugs and heated travel cups typically divide their production into two major stages which are structural fabrication and final electronic assembly. As according to (KingStar 2019) and (Diller 2019), the smart vacuum flask body usually consist of an inner and outer stainless-steel layer that is produced through deep drawing, vacuum insulation and welding processes, which are commonly outsourced to specialized stainless-steel manufacturers.



*Figure 4: Current Market Manufacturing Process Flow*

Similarly, the process for outer bottle and inner bottle which is outsourced the manufactured, the pattern and logo printing will be based on our design which the outside manufacturing company would receive. After the packing it would then be sent to our company to perform electronic component integration.



*Figure 5: In-House Smart Component Integration*

To ensure consistent product quality, efficiency and safety, our company adopts a semi-automated production system supported by robotic assembly and machine vision inspection. The focus of the process is the integration of electronic components (LED screen, thermostat, battery module, and heating element) into the pre-manufactured smart vacuum flask body.

## Testing and Inspection Equipment

| Equipment                            | Function                                                                                                                                                                                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine Vision Inspection System     | Utilizes high-resolution cameras and AI-based recognition to check component placement, solder quality, and logo alignment automatically. Detects defects that are invisible to the human eye. |
| Temperature Calibration Bench        | Automated testing platform that measures sensor accuracy and verifies heating performance within a set temperature tolerance ( $\pm 1^{\circ}\text{C}$ ).                                      |
| Battery Charging and Safety Analyzer | Tests battery charging cycles, voltage stability, and protection circuitry. Ensures compliance with electrical safety standards.                                                               |
| Vacuum Integrity Tester              | Confirms that the flask body maintains proper vacuum insulation before electronic assembly.                                                                                                    |

*Table 2: In-House Testing and Inspection Equipment*

### Facilities and Support Areas

| Equipment                    | Function                                                                                                                         |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Clean Assembly Room          | Provides dust-free and temperature-controlled conditions for electronics assembly.                                               |
| Quality Control Laboratory   | Equipped with environmental test chambers for thermal stress and lifespan testing.                                               |
| Storage and Warehouse        | Separate areas for raw materials, semi-finished goods, and finished products. Battery storage is isolated for safety compliance. |
| Packaging and Labelling Area | Automated labelling line prints QR codes, logos, and production date before packaging.                                           |

*Table 3: In-House Facilities and Support Areas*

The company plans to manufacture and distribute approximately 37,500 units of the smart vacuum flask per year, targeting both local and regional consumer markets. This annual production goal represents a moderate to high volume output, suitable for a consumer product with steady demand. The product variety remains low to moderate, consisting mainly of variations in external colour, logo design and display customization, while maintaining the same internal electronic configuration. Based on this volume variety balance, the company adopts a product focused strategy, supported by a manual labour assembly- line layout for production.

This setup allows workers to efficiently integrate the LED screen, thermostat, battery and heating element in a consistent sequence, minimizing setup time between product variants. Although the assembly process is manual, automated inspection systems are utilized for quality checking to ensure uniform standards across all units. The chosen approach allows the company to achieve economies of scale while maintaining enough flexibility to incorporate minor design changes or product updates. This ensures the company can meet its 37,500 units annual production target while sustaining competitiveness and reliable product performance.

## 7.3 Characteristics of the Location

The location of operation for our business should consist of excellent logistics connectivity, access to a skilled workforce, robust utilities and infrastructure, cost-effective operations and a supportive regulatory environment. These criteria are vital in determining a suitable and strategic site for our company's assembly and distribution activities, ensuring smooth daily operations and long-term growth.

### 7.3.1 Logistics Connectivity

An ideal location allows seamless coordination between our design and OEM manufacturing partners, minimizes transportation costs and provides quick access to both domestic and international markets. For instance, Malaysia's industrial hubs such as Penang are well-known for their world-class logistics networks that integrate seaports, airports and highways, enabling efficient export and import activities crucial for medium-sized manufacturers and assembly firms (MIDA 2025).

The logistics industry in Penang is supported by the Penang Port, one of the busiest ports in Malaysia, which facilitates the handling of millions of tons of cargo annually. This strategic advantage underscores the significance of logistics companies in Penang, as they leverage the port to streamline shipping operations and create efficient links to major trade routes (Penang Port, Malaysia 2025).

#### **Key Players in Penang's Logistics Sector**

Several prominent logistics companies operate in Penang, offering diverse services that cater to different business needs. These companies range from global giants to local enterprises, each contributing to the robustness of the logistics sector in Penang.

##### **1. TNLHS Logistics**

As a leading logistic company in Penang, TNLHS Logistics provides comprehensive solutions that include freight forwarding, warehousing and customs brokerage. Their deep understanding of the local market and international logistics makes them a preferred partner for businesses looking to optimize their supply chains (TNL 2024).

##### **2. DHL Express**

A global leader in express services, DHL has a strong presence in Penang. Their extensive network and advanced technology ensure fast and reliable delivery services, both domestically and internationally, making them a key player in maintaining high standards in the region (DHL 2025).

##### **3. Omni Logistics**

Omni Logistics is a reputable logistics company strategically located within the Penang Free Trade Zone, offering a full range of supply chain solutions tailored to manufacturing and assembly businesses. The company specializes in freight forwarding, warehousing and customs clearance, enabling smooth coordination between suppliers, OEM manufacturers and our company. With its

advanced tracking systems and strong expertise in handling industrial and electronic products, Omni Logistics ensures timely and secure transportation of both raw materials and finished goods (Omni Logistics 2024).

## 7.4 Labour Availability and Skills

When selecting a factory site for ThermaCorex's vacuum flask assembly operations, access to a skilled and semi-skilled workforce is one of the most critical factors to ensure production efficiency and quality. Penang, particularly the Bayan Lepas Free Industrial Zone, offers a competitive advantage due to its concentration of technically trained workers experienced in manufacturing, electronics assembly and precision engineering. The presence of training institutions such as the Penang Skills Development Centre (PSDC) and Universiti Sains Malaysia (USM) ensures a continuous pipeline of skilled graduates and technicians who can meet evolving industry demands. Labour costs in Penang remain relatively moderate compared to other developed industrial regions in Asia, allowing ThermaCorex to maintain cost-effective operations without compromising on expertise (Nasir 2025). Furthermore, the state government's initiatives to enhance workforce readiness and promote Industry 4.0 adoption help businesses access upskilling programs and technical certifications (MIDA 2025).

## 7.5 Utilities & infrastructure

Penang offers highly reliable and modern infrastructure, making it an ideal location for ThermaCorex's vacuum flask assembly facility. The state provides stable electricity and water supply through Tenaga Nasional Berhad (TNB) and Perbadanan Bekalan Air Pulau Pinang (PBAPP), both known for consistent service and low downtime (PBAPP 2020). Waste disposal and recycling services are well-managed under Penang's Green State initiative, promoting sustainability and compliance with environmental standards (PGC 2016). High-speed broadband and telecommunications infrastructure are readily available, supported by major providers such as TM and Maxis, ensuring seamless communication and integration of digital systems essential for Just-In-Time (JIT) operations and supplier coordination. The presence of fully serviced industrial zones, such as Bayan Lepas Free Industrial Zone, offers comprehensive factory facilities, including logistics access, maintenance services and utilities management.

## 7.6 Cost

Operating in Penang allows ThermaCorex to benefit from competitive cost structures while maintaining high operational standards. The capital cost of factory rental within the Bayan Lepas Industrial Zone remains moderate compared to major industrial cities in Asia, making it suitable for medium-scale startups. Utility costs, including electricity and water, are relatively low and stable, supported by Malaysia's government-regulated pricing system. Additionally, businesses within the Free Industrial Zone (FIZ) enjoy tax incentives, import duty exemptions and streamlined customs procedures, effectively reducing overhead and improving profit margins (Asian

Development Bank 2021). These cost advantages allow ThermaCorex to maintain competitive pricing for its products while investing more in innovation and quality assurance.

## 7.7 Potential Factory Site

In selecting a suitable location for ThermaCorex’s assembly facility, Penang emerges as the most strategic region due to its well-established industrial ecosystem, advanced infrastructure, strong logistics connectivity and supportive policies for manufacturing and export-oriented SMEs. The state hosts a dense concentration of electronics, precision engineering and supply-chain service providers, making it highly compatible with the production requirements of smart vacuum flask manufacturing operation. In addition, Penang’s geographic position which is situated near key global shipping routes and served by Penang International Airport offers significant advantages in terms of import efficiency, outbound distribution and integration with OEM suppliers.

To systematically evaluate the most suitable site within Penang, the factor-rating method was applied, allowing multiple criteria to be compared using weighted scores aligned with the operational needs of the business. Three of Penang’s most established industrial areas which are Bayan Lepas Free Industrial Zone, Bukit Minyak Industrial Park and Batu Kawan Industrial Park were shortlisted due to their accessibility and relevance for light assembly operations. Each location was assessed based on critical factors such as rental cost, logistics accessibility, supplier proximity, skilled labour availability and tax structure. This structured approach ensures an objective and data-driven decision, enabling the company to select a location that optimises both operational efficiency and long-term scalability.

| Factor                      | Weight | Scores (Out of 100) |              |            | Weighted Scores |              |              |
|-----------------------------|--------|---------------------|--------------|------------|-----------------|--------------|--------------|
|                             |        | Bayan Lepas         | Bukit Minyak | Batu Kawan | Bayan Lepas     | Bukit Minyak | Batu Kawan   |
| Rental Cost                 | 0.25   | 70                  | 90           | 80         | 17.5            | 22.5         | 20           |
| Logistics Accessibility     | 0.20   | 95                  | 80           | 85         | 19              | 16           | 17           |
| Supplier Proximity          | 0.10   | 95                  | 80           | 75         | 9.5             | 8            | 7.5          |
| Skilled Labour Availability | 0.25   | 95                  | 85           | 85         | 23.75           | 21.25        | 21.25        |
| Tax Structure               | 0.20   | 85                  | 75           | 85         | 17              | 15           | 17           |
| Totals                      | 1      |                     |              |            | <b>86.75</b>    | <b>82.75</b> | <b>82.75</b> |

Table 4: Factor Rating Method

Based on the weighted scores generated from the factor-rating Method, Bayan Lepas Free Industrial Zone achieved the highest overall score, indicating that it is the most suitable location for ThermaCorex’s assembly operations. Although its rental cost is comparatively higher, this

disadvantage is outweighed by its superior logistics accessibility, strong supplier networks, availability of skilled labour and supportive tax structures for export-oriented activities. Although Bukit Minyak and Batu Kawan also demonstrated strong performance, their relative distance from major logistics hubs and supplier clusters places them at a slight disadvantage compared to Bayan Lepas. Overall, the analysis demonstrates that Bayan Lepas provides the most balanced combination of operational efficiency, supply-chain integration and long-term scalability.

In conclusion, the systematic evaluation conducted through the factor-rating method clearly identifies Bayan Lepas Free Industrial Zone as the optimal choice for hosting ThermaCorex's factory site. Its Free Trade Zone benefits, skilled workforce and superior logistics infrastructure align closely with the company's export-oriented objectives.

The Bayan Lepas Free Industrial Zone (BLFIZ), established in 1972 as Malaysia's first Free Industrial Zone, is globally recognized for driving Penang's industrial growth and attracting foreign investment (Asian Development Bank 2021). Additionally, its Free Trade Zone (FTZ) status provides tax exemptions on imported raw materials, machinery and components which is a key advantage that reduces production costs and enhances export competitiveness (Asian Development Bank 2021). The zone also offers excellent logistics connectivity, being located near the Penang International Airport, major seaports and bridges linking to the mainland, making it highly suitable for export-oriented industries.

Penang additionally offers a strong manufacturing ecosystem and skilled workforce. The Penang Skills Development Centre (PSDC), located within the Bayan Lepas area, serves as a leading training hub that supplies skilled and semi-skilled labour for the state's high-tech and manufacturing sectors. A recent study by the Penang Institute (2023) confirmed the rising demand for engineering and technical talent, underscoring the strength of Penang's industrial labour base (Penang Institute, 2023).

While BLFIZ provides significant operational and logistical advantages, it also presents certain challenges such as higher land and labour costs due to competition from established multinational manufacturers, as well as traffic congestion and limited land availability on the island. Nevertheless, the Penang State Government has assured that there will be sufficient industrial land for at least the next 10–15 years, mitigating long-term expansion risks.

| Location                                                | Advantages                                                                                                                                                                                                              | Disadvantages                                                                                                                                                                                         |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bayan Lepas Free Industrial Zone (BLFIZ), Penang Island | <ul style="list-style-type: none"> <li>• Free Trade Zone (FTZ) status offers tax exemptions on imported raw materials, machinery and components.</li> <li>• Excellent logistics connectivity close to Penang</li> </ul> | <ul style="list-style-type: none"> <li>• Higher land and property costs due to high demand and limited space on the island.</li> <li>• Labour competition with high-tech and semiconductor</li> </ul> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                          |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>International Airport, seaports and bridge links to the mainland.</p> <ul style="list-style-type: none"> <li>• Established industrial ecosystem with strong supplier networks, machining and assembly support.</li> <li>• Abundant skilled and semi-skilled labour from Penang's manufacturing sector, supported by training centres such as the Penang Skills Development Centre (PSDC).</li> <li>• Strong government and SME support via InvestPenang and the Penang SME Centre.</li> <li>• Assured industrial land pipeline for 10–15 years, reducing risk of land shortage.</li> </ul> | <p>firms may increase wages or cause shortages.</p> <ul style="list-style-type: none"> <li>• Traffic congestion and commuting challenges, especially during peak hours.</li> <li>• Higher living costs for workers, which may affect labour retention and operating expenses.</li> </ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 5: Advantages and Disadvantages of Bayan Lepas

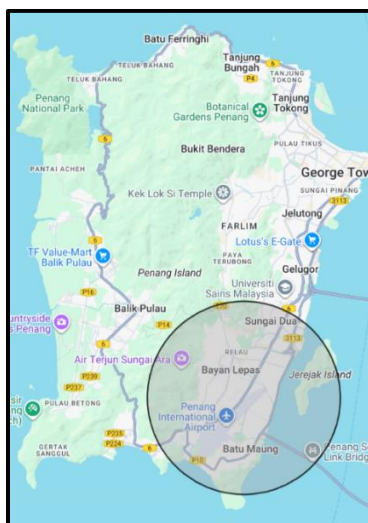


Figure 6: Proposed Location for ThermoCorex

## 7.8 Facility Layout & Blueprint

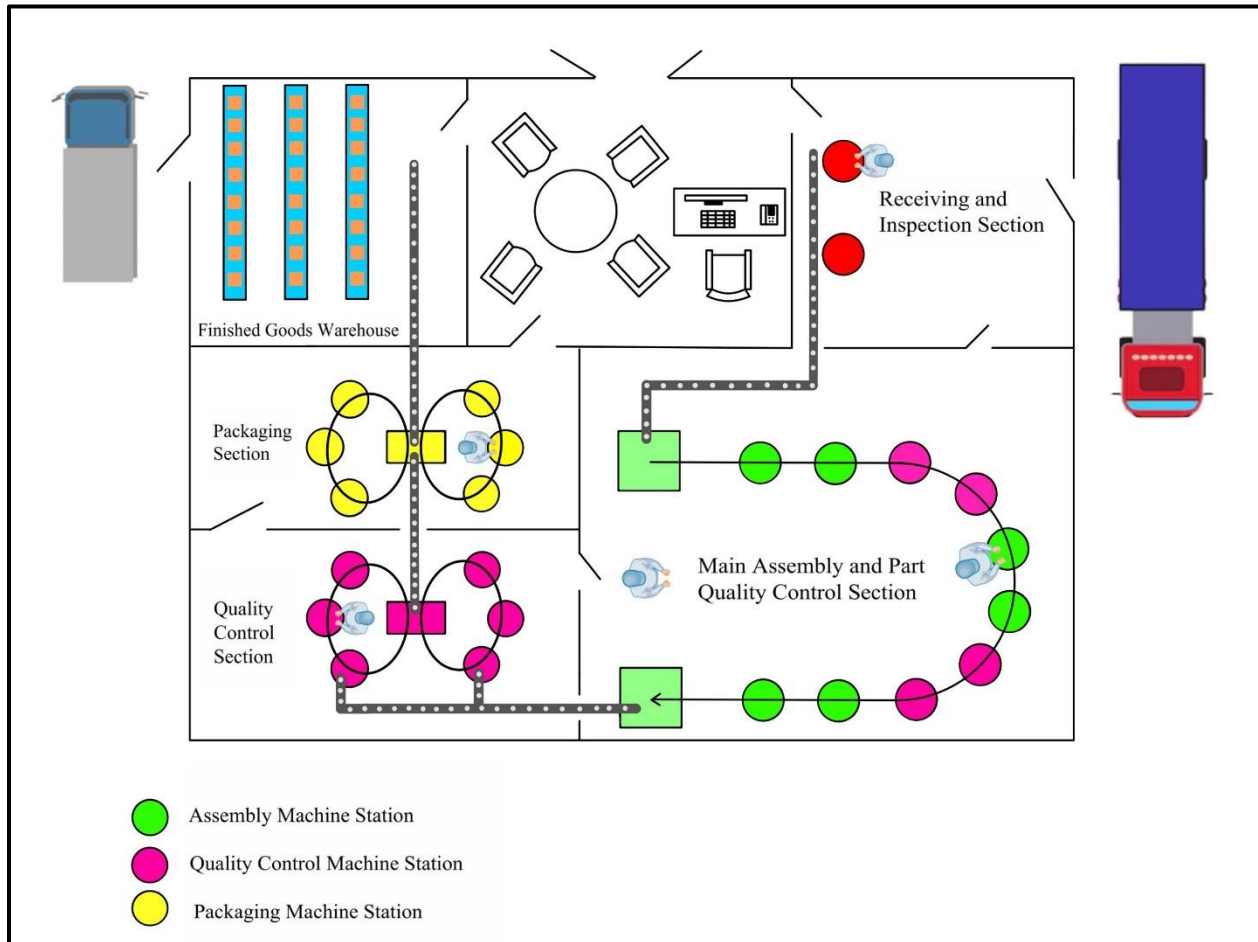


Figure 7: Proposed Layout of ThermaCorex

Since our business is product-oriented and currently focuses on producing a single model of smart vacuum flask, our facility layout is designed for efficiency, compactness and smooth workflow continuity. Due to the high rental costs within the Bayan Lepas Industrial Zone, our operation is housed in a limited yet well-organized facility that maximizes space utilization while ensuring an uninterrupted flow of production activities. The factory adopts a product-oriented (line flow) layout, ideal for standardized and repetitive production processes that follow a fixed sequence of operations.

The layout, as illustrated in Figure 5, is designed to provide a linear flow of materials from receiving to dispatch, minimizing unnecessary handling and transportation. Key operational zones include the Receiving and Inspection Area, Main Assembly Line, Quality Control (QC), Packaging Section, Finished Goods Warehouse and Dispatch Area, supported by an office and administrative space centrally located for production supervision.

$$\begin{aligned}
 \text{Takt time} &= \frac{\text{Total time available}}{\text{Units required}} \\
 &= \frac{720}{125} \\
 &= 5.76 \text{ mins/unit} \\
 \\ 
 \text{Workers required} &= \frac{\text{Total operation time required}}{\text{Takt time}} \\
 &= \frac{25.2}{5.76} \\
 &= 4.375 \\
 &\approx 5 \text{ workers}
 \end{aligned}$$

Based on a total operation time required which is 0.42 hours per unit (25.2 minutes) and a daily customer demand of 125 units, the calculated takt time is 5.76 minutes per unit. This means the production line must complete one unit approximately every 5.8 minutes to satisfy demand. Dividing the total work content by the takt time indicates that 4.38 workers are required, which is rounded up to five cross-trained operators. This ensures that all tasks within the work cell can be completed without delays, maintaining a continuous production flow aligned with customer demand.

The production process begins at the Receiving and Inspection Area, where stainless-steel vacuum flask shells, electronic display modules, batteries, silicone seals and packaging materials are unloaded and inspected. Once verified, materials are moved to the Main Assembly Line, which forms the core of the production process. This area features a U-shaped work cell layout, allowing workers to access multiple assembly points within a compact footprint. The U-shape is particularly suitable for small-scale facilities like ours, as it reduces employee movement, minimizes space requirements and enhances communication among workers. This configuration enables operators to assemble components efficiently from soldering electronic temperature modules to fitting seals and attaching flask caps while facilitating easier in-process inspection and reducing the total number of workers required without compromising productivity. The close proximity of workstations within the U-shaped line also promotes collaboration and quick problem-solving during production.

After assembly, the vacuum flasks are transferred to the Quality Control (QC) Section, which is arranged as a dedicated small work cell within a confined area of the facility. Due to limited space availability, the QC layout is designed for compactness and efficiency, with testing stations positioned back-to-back. Each workstation is equipped with leak testers, display function testers and visual inspection tools to ensure product quality meets design specifications. The layout allows

inspectors to work in close coordination and share testing equipment conveniently, reducing idle time and optimizing use of space.

Next, the flasks proceed to the Packaging Section, located adjacent to the QC area. Similar to the QC layout, the packaging section is compactly arranged to fit within the small, enclosed area available. The layout includes two side-by-side packaging tables, allowing workers to label, box and seal units efficiently in pairs. This arrangement minimizes walking distance and handling time, supporting a Just-In-Time (JIT) production approach where products are packed immediately after inspection. The efficient use of limited space ensures that even with high output, workflow remains smooth without congestion or delays.

Packaged units are then moved to the Finished Goods Warehouse, where products are stored temporarily on pallet racks and shelves before being shipped. The Dispatch Area, located near the loading dock, allows easy truck access for outbound logistics handled by partners. Finally, the office space is strategically positioned in the centre of the facility, giving supervisors a clear view of the production floor for effective coordination and operational control.

Overall, the product-oriented layout ensures a continuous and logical flow of materials and products, with each section arranged to optimize efficiency and quality control within a space-constrained environment. The U-shaped main assembly work cell enhances worker productivity, communication and inspection efficiency, while the compact QC and packaging layouts make effective use of limited floor area. This configuration aligns with lean manufacturing principles, reducing waste, minimizing movement and ensuring high product quality while keeping operating costs low. The design also provides flexibility for future expansion as production volume increases or new vacuum flask models are introduced.

## 7.9 Inventory & Purchasing

As our business focuses on the assembly and packaging of smart vacuum flasks, most of the core components are sourced from external suppliers and OEM manufacturers. The inventory we manage primarily consists of electronic, sealing and packaging parts, along with the stainless-steel vacuum flask shells produced by our OEM partner.

To ensure the consistent production of premium-quality vacuum flasks, ThermaCorex places strong emphasis on sourcing materials and components exclusively from reputable and well-established OEM manufacturers within the global drinkware and thermal insulation industry. These partnerships are fundamental to maintaining the company's product integrity, durability and performance standards. The global supply chain for stainless-steel and vacuum-insulated containers is highly developed, particularly in industrial hubs such as China and Japan, where manufacturing technology and quality control practices have reached world-class levels. Manufacturers in these countries have extensive experience in precision metal forming, vacuum sealing, and double-wall insulation processes, enabling them to produce highly efficient and

durable flask shells that meet international safety and food-grade standards such as CE, FDA, LFGB and RoHS certifications.

To secure dependable and high-quality vacuum flask components, ThermaCorex carefully evaluates potential OEM partners based on their production capabilities, quality assurance standards and supply chain practicality. One potential supplier is Suzhou Prime Group, a China-based manufacturer located in Jiangsu Province. With over ten years of experience in the drinkware manufacturing industry, Suzhou Prime Group is recognized as one of China's leading producers of stainless-steel vacuum flasks and water bottles. The company complies with international safety and quality certifications such as CE, FDA, LFGB and RoHS, ensuring that its products meet American and European food-contact standards. Its large-scale production capacity and competitive pricing make it attractive for mass production. However, due to its geographical distance, sourcing from China introduces several challenges for ThermaCorex, including higher shipping costs, longer lead times, customs procedures and potential delays caused by cross-border logistics or global supply chain disruptions. These factors make Suzhou Prime Group less suitable as a primary supplier during ThermaCorex's initial operational phase.

Another notable manufacturer is Zojirushi Corporation of Japan, a globally renowned leader in precision-engineered thermal products. Founded in 1918, Zojirushi specializes in vacuum-insulated flasks, beverage dispensers and household appliances such as rice cookers and electric water boilers. With subsidiaries in Taiwan, China, Hong Kong, South Korea, and the United States, the company is recognized for its cutting-edge vacuum insulation technology and superior product quality (Zojirushi 2025). Despite its global reputation, Zojirushi primarily serves as a consumer brand manufacturer rather than a contract OEM. Moreover, its high production costs and long-distance logistics make it impractical for ThermaCorex's startup scale. In addition, international transactions involving currency exchange and complex payment systems may further increase financial risk and administrative burdens. Therefore, while Zojirushi sets an industry benchmark in product excellence, it is not a feasible supplier for ThermaCorex's current operational model.

In contrast, focusing on regional manufacturers such as those in Malaysia, Singapore, and Thailand offers a more strategic and sustainable approach for ThermaCorex's supply chain. Local and regional sourcing provides shorter lead times, lower transportation costs, easier communication and better adaptability to changing market demands. Vacuum Tech Manufacturing (Malaysia) Sdn. Bhd located in Shah Alam, Selangor is a domestic manufacturer specializing in stainless-steel vacuum flasks. This company provides efficient production schedules and reliable after-sales support. Working with local manufacturers also reduces environmental impact through lower carbon emissions and supports Malaysia's Small and Medium Enterprise (SME) ecosystem, promoting local employment and industrial growth. The following table presents a comparative summary of the three suppliers evaluated.

| <b>Criteria</b>                           | <b>Suzhou Prime Group</b> | <b>Zojirushi Corporation</b> | <b>Vacuum Tech Manufacturing (Malaysia) Sdn. Bhd.</b> | <b>Explanation</b>                                                                                                                                     |
|-------------------------------------------|---------------------------|------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supplier Stability and Reliability        | 5                         | 5                            | 5                                                     | Vacuum Tech Manufacturing is a highly reliable and established producer, serving as an OEM manufacturer for the renowned vacuum flask brand 'Thermos'. |
| Technical Support and After-Sales Service | 4                         | 3                            | 5                                                     | Located locally, Vacuum Tech provides faster and more effective technical support and after-sales service, with no language or communication barriers. |
| Scalability of Supply                     | 5                         | 5                            | 4                                                     | Suzhou Prime Group and Zojirushi offer large-scale production, while Vacuum Tech's flexible capacity is ideal for start-ups and medium enterprises.    |

|                              |   |   |   |                                                                                                                                                    |
|------------------------------|---|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Safety and quality Standards | 5 | 5 | 5 | All three companies comply with international safety, quality, and environmental standards such as CE, FDA, LFGB, and ISO certifications.          |
| Pricing and Payment Options  | 3 | 3 | 5 | Vacuum Tech offers cost-effective pricing and convenient local payment terms in MYR, avoiding international currency conversion and transfer fees. |
| Proximity and Logistics      | 3 | 3 | 5 | Being locally based, Vacuum Tech ensures shorter lead times, lower transportation costs, and quicker delivery compared to overseas suppliers.      |

*Table 8: Comparative Summary of Suppliers*

From the supplier evaluation table above, Vacuum Tech Manufacturing (Malaysia) Sdn. Bhd. achieved the highest overall evaluation scores.

### **Supplier Strategy and Competitive Advantage Alignment**

ThermaCorex adopts a differentiation strategy, positioning itself as a provider of premium, high-quality smart vacuum flasks that offer enhanced durability, advanced electronic features and functional design. Under this strategy, the main source of competitive advantage comes from product uniqueness, exceptional quality and enhanced user experience, rather than competing purely on cost. Therefore, the supplier strategy must focus on securing partners capable of consistently delivering components that support this higher level of product performance.

Selecting Vacuum Tech Manufacturing (Malaysia) Sdn. Bhd. as the primary OEM supplier strongly aligns with this differentiation approach. As a reputable Malaysian manufacturer experienced in producing stainless-steel vacuum flasks including serving as an OEM for the well-known ‘Thermos’ brand, Vacuum Tech provides components that meet stringent quality standards such as CE, FDA, LFGB and ISO certifications. Its proven expertise in double-wall insulation, vacuum sealing precision and stainless-steel forming ensures that ThermaCorex’s core product maintains superior thermal performance and structural durability, which are essential traits for differentiated products. Being located within Malaysia also enables tighter quality oversight, frequent inspections and close collaboration on product improvements, all of which enhance product refinement and support continuous innovation.

Moreover, establishing strong partnerships with nearby manufacturers significantly enhances supply chain resilience, particularly for a startup like ThermaCorex that values stability and agility during its early growth phase. Local suppliers are inherently more responsive to fluctuations in order volume, design modifications and quality improvement feedback, enabling the company to refine and optimize its products more efficiently. Investing in local sourcing not only reduces logistics costs but also supports a more sustainable, community-driven business model. Regional procurement fosters operational flexibility while strengthening local industry ecosystems (Das 2023). Therefore, prioritizing partnerships with nearby manufacturers such as Vacuum Tech Manufacturing (Malaysia) Sdn. Bhd. provides ThermaCorex with a strategic balance between cost efficiency, adaptability and environmental responsibility. This localized sourcing approach establishes a solid foundation for gradual expansion and prepares the company for future collaboration with international suppliers as production volumes and market demand increase.

The table below shows the supplier chosen for individual electronic components, packaging materials and adhesive.

| Inventory             | Description                                                                                                             | Supplier                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Printed Circuit Board | Custom-designed printed circuit board used to integrate all the electronic components together with the microcontroller | Elna PCB (M) Sdn. Bhd.                                 |
| Battery               | Compact and long-lasting battery designed for low-power electronics.                                                    | Innovate New Materials Technology (Malaysia) Sdn. Bhd. |

|                         |                                                                                                                                                         |                                    |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Adhesive and Screws     | Industrial-grade adhesive, precision micro screws and fasteners used to secure the display module, internal PCB, and flask shell during assembly.       | Fon Kok Fastener Sdn. Bhd.         |
| Packaging Box and Label | Custom eco-friendly packaging boxes made from recyclable corrugated cardboard, along with printed product labels and brand tags to protect the flask.   | Hung Print Industries Sdn. Bhd     |
| Electronic Components   | Includes tiny electronic components such as sensors, microcontroller and LED display modules integrated within the vacuum flask's smart display system. | Guan Seng Hing Electronics Sdn Bhd |

*Table 9: Chosen Supplier per component*

ThermaCorex adopts a response strategy by prioritising speed, flexibility and the ability to react rapidly to fluctuations in customer demand. This aligns with the company's competitive advantage of delivering locally assembled smart vacuum flasks with shorter lead times, reliable availability and fast after-sales support. By sourcing electronic components, packaging materials, adhesives and fasteners from local Penang-based suppliers, ThermaCorex's supply chain responsiveness is enhanced. Local sourcing reduces lead time variability, minimizes disruptions and enables just-in-time (JIT) inventory practices, reducing the need for high buffer stocks. When customer demand increases, design changes occur or quality issues arise, nearby suppliers allow the company to react immediately, preventing downtime and maintaining smooth assembly operations.

The combined effect of localized sourcing is a highly flexible supply chain capable of adapting to market conditions and production changes with minimal delay. This ensures that ThermaCorex maintains a strong competitive edge by offering faster delivery, consistent product quality and agile operational performance.

## 7.10 Purchase Policy

A well-structured purchasing policy plays a crucial role in ensuring operational efficiency, effective cost management and alignment with ThermaCorex's strategic goals as an emerging and sustainability-oriented enterprise. To maintain uninterrupted production while avoiding excessive inventory costs, the company adopts a purchasing framework built around three key principles which are Just-in-Time (JIT) procurement and ABC inventory classification. These strategies collectively enable ThermaCorex to balance supply reliability with financial prudence, ensuring that components and materials are procured at the right time, in the right quantity and from dependable sources.

### **Just-in-Time (JIT) Procurement**

Just-in-Time procurement refers to receiving goods from suppliers precisely when they are needed in the production process, rather than holding excessive stock. The main objective of JIT is to reduce inventory holding costs and increase turnover (Taghipour 2019). For ThermaCorex, adopting a JIT strategy means ordering components such as vacuum-flask shells, battery, PCB, and electronic components close to the point of assembly, thereby reducing capital tied up in storage and minimizing obsolete inventory risk. This approach significantly reduces inventory costs by lowering storage requirements, minimizing waste and eliminating the need for large warehouses. It also improves efficiency as production schedules are closely aligned with component delivery, reducing idle time and ensuring smoother workflow within the assembly line. Furthermore, JIT enhances flexibility, allowing ThermaCorex to quickly adjust production levels according to demand changes without overstocking or shortages. Overall, this system reduces operational costs by optimizing resource utilization, improving cash flow, and streamlining coordination between suppliers and production. However, this strategy demands highly reliable supplier networks and precise demand forecasting since disruptions may lead to production stoppages (Reynolds 2025). To mitigate these risks, ThermaCorex will maintain a small safety stock for critical components and work with local manufacturers for faster lead times and responsive logistics. The process of JIT inventory management is summarized in the picture below.

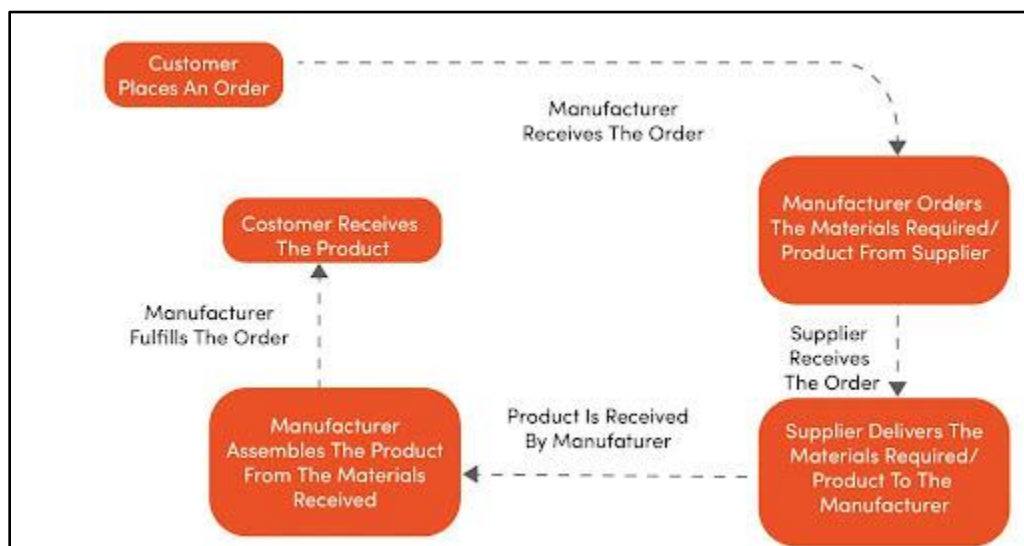


Figure 7: JIT Inventory Management (Sk 2021)

### **ABC Inventory Classification**

ABC classification segments inventory into three categories based on consumption value and importance. Class A items such as stainless-steel vacuum flask shells are high-value materials that serve as the core components of production and represent a significant portion of total

manufacturing costs. These items require strict monitoring, precise quantity tracking and tight inventory control to avoid shortages or excess stock that could disrupt the assembly process. Class B items including batteries, printed circuit boards (PCBs) and other electronic components, hold moderate value and are essential for the functionality of the smart vacuum flask. They are reviewed periodically to maintain optimal stock levels and ensure continuous supply without tying up unnecessary capital. Class C items such as packaging materials, adhesives and screws are low-cost consumables that can be purchased in bulk and managed with basic record-keeping due to their low financial impact.

| Type           | Importance          | Percentage of Total Inventory | Annual Consumption Value | Controls | Records       |
|----------------|---------------------|-------------------------------|--------------------------|----------|---------------|
| <b>Class A</b> | High dollar value   | 10% – 20%                     | 70% – 80%                | Tight    | High Accuracy |
| <b>Class B</b> | Medium dollar value | 30%                           | 15% – 20%                | Medium   | Good          |
| <b>Class C</b> | Low dollar value    | 50%                           | 5%                       | Basic    | Minimal       |

*Table 10: ABC Inventory Classification*

By applying this classification approach, ThermaCorex ensures that management of attention and financial resources are directed toward materials that most influence production continuity and cost efficiency. This system enhances working capital optimization, reduces the risk of overstocking and ensures that critical components remain readily available for smooth and uninterrupted production operations.

## 7.11 Economic Order Quantity (EOQ)

The Economic Order Quantity (EOQ) model is a fundamental inventory management tool used by ThermaCorex to determine the most cost-effective quantity of materials to order at one time. Its objective is to minimize the total cost of inventory by balancing two key expenses: ordering costs (the cost of placing and receiving orders) and holding costs (the cost of storing inventory over time). By finding the optimal order size, ThermaCorex can ensure that inventory is replenished efficiently - avoiding both shortages and excessive stock.

The EOQ formula is expressed as:

$$Q^* = \sqrt{\frac{2DS}{H}}$$

Where:

D = Annual demand (units per year)

S = Ordering cost per order (RM)

H = Holding cost per unit per year (RM)

Given the values:

$$H = RM23.75$$

$$D = 37500 \text{ units}$$

$$S = RM100/\text{order}$$

To calculate the optimal number of pieces per order,  $Q^*$ :

$$\begin{aligned} Q^* &= \sqrt{\frac{2(37500)(100)}{23.75}} \\ &= 561.95 \\ &\approx 562 \text{ units/order} \end{aligned}$$

For cost-effective production, ThermaCorex should order approximately 562 vacuum flask shells per batch from Vacuum Tech company. This order quantity minimizes the combined cost of ordering and storing inventory, helping the company to reduce warehouse space usage and storage cost. Besides that, implementing this model also maintain smooth production flow without overstocking, improve cash flow by preventing excessive capital lock-up in inventory and align procurement schedules with Just-In-Time (JIT) principles and supplier lead times.

## 7.12 Warehouse and Storage Requirements

To accommodate our final products for a short period of time, establishing an efficient and well-structured warehouse is essential to support smooth assembly, storage and distribution of vacuum flasks. Since the business operates on a Just-in-Time (JIT) and product-oriented layout, the warehouse must balance adequate storage capacity with efficient material flow. The warehouse must be a part of our operating place to store finished vacuum flasks temporarily. Given the limited floor space in our facility, the warehouse is designed for compact vertical storage with FIFO (First-In, First-Out) inventory management to ensure smooth material rotation and reduce aging stock.

## 8.0 Sustainability Issue

### 8.1 Sustainability Strategy

This section outlines the sustainability strategy implemented for the ThermaCorex Smart Vacuum Flask, focusing on the product's life cycle assessment (LCA), waste and hazard management and principles of recycling, reworking and reuse. The aim is to optimize environmental performance and resource efficiency while maintaining durability, safety, and consumer satisfaction. The integration of digital manufacturing technologies and automation further enhances production consistency and reduces environmental footprint.



**Model Name:** Uma ThermoFlask

**Weight:** 1290.11 g

**Built to last:** 5.0 year

**Duration of use:** 5.0 year



**Manufacturing Region**

The choice of manufacturing region determines the energy sources and technologies used in the modeled material creation and manufacturing steps of the product's life cycle.

**Use Region**

The use region is used to determine the energy sources consumed during the product's use phase (if applicable) and the destination for the product at its end-of-life. Together with the manufacturing region, the use region is also used to estimate the environmental impacts associated with transporting the product from its manufacturing location to its use location.

## Sustainability Report

Model Name: Uma ThermoFlask

Weight: 1290.11 g

Built to last: 5.0 year

Duration of use: 5.0 year

### Assembly Process

Region: Asia  
Energy type: Electricity  
Energy amount: 0.05 kWh  
Built to last: 5.0 year

### Use

Region: Asia  
Energy type: None  
Energy amount: 0.00 kWh  
Duration of use: 5.0 year

### Transportation

Truck distance: 17 km  
Train distance: 0.00 km  
Ship distance: 6700 km  
Airplane Distance: 6700 km

### End of Life

Recycled: 33 %  
Incinerated: 13 %  
Landfill: 54 %

Model Name: Uma ThermoElask

Weight: 1290.11 g

Built to last: 5.0 year

Duration of use: 5.0 year

### Environmental Impact (calculated using CML impact assessment methodology)

#### Carbon Footprint



2600 kg CO<sub>2</sub>e

|                 |                            |
|-----------------|----------------------------|
| Material:       | 32 kg CO <sub>2</sub> e    |
| Manufacturing:  | 2600 kg CO <sub>2</sub> e  |
| Use:            | 0.00 kg CO <sub>2</sub> e  |
| Transportation: | 10.0 kg CO <sub>2</sub> e  |
| End of Life:    | 0.710 kg CO <sub>2</sub> e |

#### Total Energy Consumed



4.2E+4 MJ

|                 |           |
|-----------------|-----------|
| Material:       | 340 MJ    |
| Manufacturing:  | 4.1E+4 MJ |
| Use:            | 0.00 MJ   |
| Transportation: | 130 MJ    |
| End of Life:    | 0.521 MJ  |

#### Air Acidification



3.3 kg SO<sub>2</sub>e

|                 |                             |
|-----------------|-----------------------------|
| Material:       | 0.117 kg SO <sub>2</sub> e  |
| Manufacturing:  | 3.1 kg SO <sub>2</sub> e    |
| Use:            | 0.00 kg SO <sub>2</sub> e   |
| Transportation: | 0.035 kg SO <sub>2</sub> e  |
| End of Life:    | 3.6E-4 kg SO <sub>2</sub> e |

#### Water Eutrophication



0.462 kg PO<sub>4</sub>e

|                 |                             |
|-----------------|-----------------------------|
| Material:       | 0.133 kg PO <sub>4</sub> e  |
| Manufacturing:  | 0.322 kg PO <sub>4</sub> e  |
| Use:            | 0.00 kg PO <sub>4</sub> e   |
| Transportation: | 6.5E-3 kg PO <sub>4</sub> e |
| End of Life:    | 8.9E-4 kg PO <sub>4</sub> e |

## Sustainability Report

Model Name: Uma ThermoFlask

Weight: 1290.11 g

Built to last: 5.0 year

Duration of use: 5.0 year

### Component Environmental Impact

Top Ten Components Contributing Most to the Four Areas of Environmental Impact

| Component             | Carbon                                                                                | Water                                                                                   | Air                                                                                      | Energy                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| ThermoFlask Body      | 340  | 0.061  | 0.435  | 5500  |
| ThermoFlask Top       | 91   | 0.016  | 0.115  | 1500  |
| ThermoFlask Base      | 83   | 0.015  | 0.104  | 1300  |
| Thermoflask drink Cap | 0.030                                                                                 | 1.5E-5                                                                                  | 7.5E-5                                                                                   | 0.490                                                                                    |
| Thermoflask steam Cap | 0.022                                                                                 | 1.1E-5                                                                                  | 5.9E-5                                                                                   | 0.358                                                                                    |

### Product Life Cycle Assessment (LCA)

The Life Cycle Assessment (LCA) framework evaluates environmental impact across all stages; material selection, production, distribution, usage, and end-of-life. The ThermaCore flask incorporates high-grade 304 recycled stainless steel and BPA-free polymer seals, ensuring long service life and minimal toxicity.

Energy consumption is reduced through automated precision forming and robotic welding, which ensure consistent quality while lowering material waste. Packaging materials are biodegradable or recyclable cardboard, minimizing landfill contributions. End-of-life disassembly is simplified to encourage metal recovery and recycling. In line with Malaysia's Low Carbon Nation Aspiration 2040, ThermaCore's LCA aligns with national goals for sustainable manufacturing, waste reduction, and renewable energy integration. Details of the life cycle sustainability approach are summarized below.

| Life Cycle Stage                              | Sustainability Focus                  | Approach & Implementation                                                                                                         | Expected Impact                                                             |
|-----------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Design &amp; Materials Selection</b>       | Eco-materials and efficiency          | Use of 30% recycled 304 stainless steels; BPA-free polymer lids; lightweight insulation core to reduce material usage.            | Reduces raw material extraction and CO <sub>2</sub> emissions.              |
| <b>Production &amp; Assembly</b>              | Energy efficiency and waste reduction | Incorporation of robot-assisted assembly for precision welding, energy-efficient furnaces, and closed-loop water cooling systems. | 20–25% reduction in energy and material waste compared to manual processes. |
| <b>Packaging &amp; Distribution</b>           | Low-impact logistics                  | Use of biodegradable cartons and optimized transport routes; compact packaging to reduce volume and shipping emissions.           | Lowers transport-related emissions and non-recyclable waste.                |
| <b>Usage Phase</b>                            | Durability and energy-free operation  | Double-wall vacuum design eliminates need for external power; digital temperature display uses ultra-low-power sensors.           | Promotes long lifespan, lowering product replacement frequency.             |
| <b>End-of-Life (Recycling &amp; Disposal)</b> | Circular economy design               | Modular design allows separation of stainless steel, electronics, and polymer components for recycling or refurbishment.          | Facilitates 80–90% material recovery at product's end-of-life.              |

*Table 11: Product Life Cycle Sustainability Approach*

## **Waste Reduction**

Manufacturing processes prioritize precision metal forming and laser cutting, minimizing offcuts and scrap. Excess stainless steel is collected and remelted in-house or returned to suppliers for reuse. Powder-coating overspray is filtered and recycled. This closed-loop practice significantly reduces landfill waste and supports a circular production ecosystem.

## **Hazard Management**

To ensure environmental and user safety, non-toxic, food-grade materials are used throughout the flask's construction. Automated systems minimize worker exposure to heat or fumes during welding. Hazardous substances such as lead, mercury, or phthalates are strictly avoided in

compliance with RoHS and REACH directives. Furthermore, manufacturing facilities adopt wastewater filtration systems and air-quality controls to reduce industrial discharge.

### **Recycle, Rework, and Reuse**

End-of-line quality checks ensure that defective or substandard products are reworked rather than discarded. Components such as lids or sensors are repaired or repurposed in secondary assembly lines. Returned flasks are disassembled, and recoverable metal parts are recycled through certified partners. The “ReCorex” recycle program encourages users to return damaged flasks for recycling in exchange for discounts, reinforcing product circularity.

### **Innovative Engineering for Sustainability**

Automation and digitalization play a central role in sustainable production. Robotic welding reduces variability and material waste, while IoT-enabled production monitoring ensures optimal resource use. Advanced Computer-Aided Manufacturing (CAM) and Finite Element Analysis (FEA) enable lightweight yet durable design optimization, minimizing overuse of materials. Additionally, the smart temperature module uses low-power microcontrollers and auto-sleep modes, extending battery lifespan and reducing electronic waste.

ThermaCorex integrates sustainability principles across every operational phase, from design to disposal. Through LCA-driven strategies, waste reduction, hazard control, and innovative automation, the product achieves both environmental responsibility and manufacturing efficiency, embodying the vision of a smart, sustainable consumer product for modern lifestyles.

## **8.2 Quality Control**

Quality control plays a critical role in ensuring that every smart vacuum flask produced meets the required performance, safety, and durability standards. In this project, quality control is integrated throughout the automated production process from material selection to final inspection to identify and prevent defects early on. To systematically identify and manage potential quality issues, a cause-and-effect diagram is used. This tool helps analyse the possible causes of product defects by categorizing them into four key factors which are materials, methods, manpower and machinery.

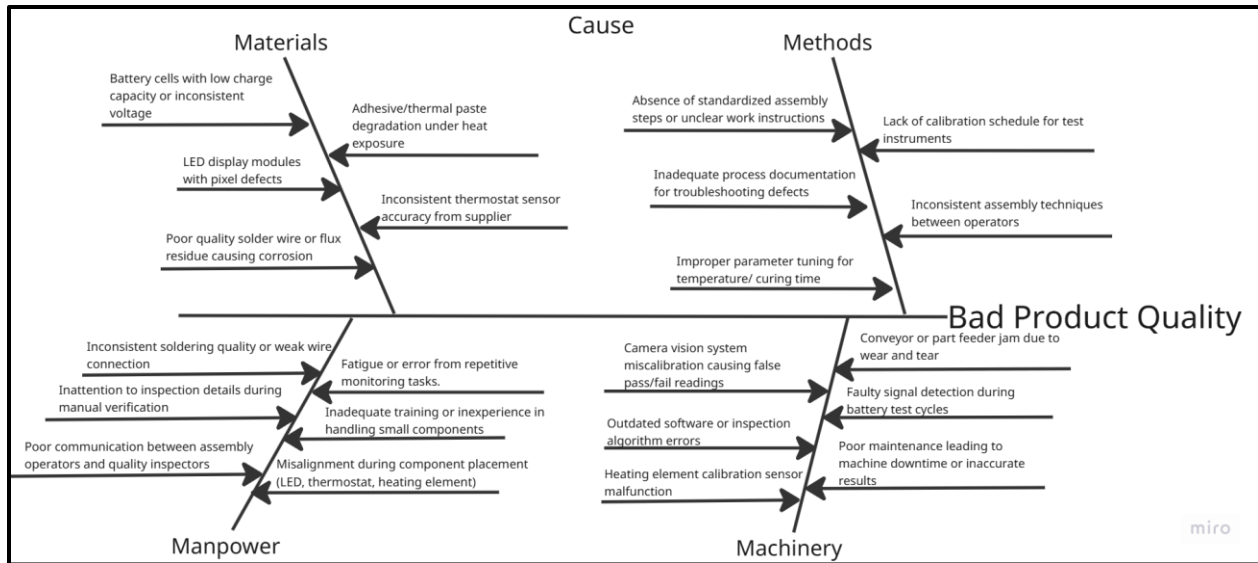


Figure 8: Cause-and-Effect During Production of Smart Vacuum Flask

Through this structured approach, the company ensures that all possible quality risks are considered and controlled. The diagram also acts as a visual reference for ongoing improvement, guiding adjustments in the automated process whenever recurring faults or deviations are identified.

In addition, to the Cause-and-Effect diagram, a Critical Control Point Table was developed to identify and monitor the possible key stages in the production process that potentially have a direct impact on product quality.

| Stage                                            | Critical Control Points                                                                         | Purpose                                                                    | Inspection Method / Tool                                              | Responsible Person / System      | Frequency   |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------|-------------|
| <b>Incoming Component Inspection</b>             | Check the conditions and specifications of LED screen, thermostat, battery and heating elements | Prevents faulty or mismatched electronic components from entering assembly | Visual inspection, multimeter test, supplier certificate verification | Incoming Quality Control team    | Every batch |
| <b>Battery &amp; Heating Element Integration</b> | Ensure proper soldering and electrical connection                                               | Prevents short circuit, overheating or power failure                       | Automated soldering inspection (AOI), continuity tester               | Assembly line manual workstation | Every unit  |

|                                             |                                                                              |                                                                 |                                                       |                                    |            |
|---------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------|------------------------------------|------------|
| <b>Temperature Sensor Calibration</b>       | Verify thermostat accuracy and display consistency                           | Ensures correct temperature reading and customer safety         | Digital thermometer comparison, calibration software  | Calibration station (automated)    | Every unit |
| <b>Vacuum Flask Body Integrity Test</b>     | Confirm outer shell's sealing and insulation before electronics installation | Prevent loss of insulation efficiency                           | Vacuum/pressure leak test, insulation resistance test | QA test system                     | Every unit |
| <b>Final Functional Test</b>                | Test LED display, heating response and battery charge cycle                  | Ensures product meets operational requirements before packaging | Full function test bench (automated)                  | End-of-line test station           | Every unit |
| <b>Aesthetic &amp; Finishing Inspection</b> | Check for scratches, logo clarity and assembly alignment                     | Maintains product appearance and brand standard                 | Visual inspection, surface scanner                    | Packaging operator / Vision system | Every unit |

*Table 12: Identification of possible Critical Control Points*

While the Cause-and-Effect diagram helps identify the potential cause of defects, the critical control point table ensure that each critical stage is monitored through specific quality control measures.

### **Customer Focus**

Beyond production, our quality control approach emphasizes customer focus as a key driver of improvement. Customer feedback, product reviews, and warranty claims are systematically collected and analysed to identify recurring issues or potential design enhancements. To achieve this a systematic online shopping platform such as: “Shopee”, “TikTok Shop” and “Lazada”, will allow customer complaints to be easily accessed and feedback to us. Moreover, social media platforms such as “Instagram” and “Facebook”, will also be used as a way to promote open discussions and reviews. This feedback loop ensures that customer satisfaction directly informs design updates and manufacturing refinements.

### **Strategic and Systematic**

Additionally, the company adopts a strategic and systematic approach aligned with strict food safety and product safety regulations, ensuring that all materials used in the smart vacuum flask

meets the health and safety standards for beverages and consumer use. Therefore, regular audits and compliance checks are conducted to uphold these standards across the supply chain.

## 9.0 Finance and Accounting Case Study

### 9.1 Business Cost

#### 9.1.1 Marketing Expenses

In our business, the business plan to market our product (smart vacuum flask) through Paid Online Advertisements and Digital Content Production.

| Paid Online Advertising                                                                                                                                                                                  | Digital Content Production                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Influencer marketing</li><li>• Sponsored posts</li><li>• Boosted posts</li><li>• Facebook/Instagram/TikTok advertisement</li><li>• Google/ Youtube Ads</li></ul> | <ul style="list-style-type: none"><li>• Graphic design (images, poster, infographics)</li><li>• Video editing</li><li>• Photography</li></ul> |

Table 13: Marketing Strategies

Using paid online advertisements and digital content production as core marketing methods is highly effective because both approaches allow businesses to precisely target audiences, enhance brand visibility, and measure performance with high accuracy. Paid advertisements provide scalable reach and flexible budget control, enabling businesses to quickly respond to market changes while driving higher conversion rates. On the other hand, digital content production strengthens engagement and supports long-term growth by delivering valuable, relevant content that resonates with potential customers over time.

In today’s marketing landscape, personalization powered by artificial intelligence, immersive technologies such as AR and VR, and conversational tools like voice assistants and chatbots are rapidly reshaping consumer expectations. These advancements reflect a broader shift toward highly tailored, interactive, and data driven marketing strategies that improve customer experience and campaign effectiveness (Arias 2024)

Additionally, complementing the effectiveness of paid ads and digital content, live streaming e-commerce has become one of the most powerful marketing trends in 2025 (Forbes Agency 2025; Forbes Business 2025; Precedence 2025). This approach merges real time video with interactive shopping, allowing brands to showcase products live, respond to customer inquiries instantly, and facilitate immediate purchases within the stream itself. Platforms such as Instagram, TikTok, and Pinterest have integrated live shopping features, making the process seamless and engaging.

Live streaming thrives because it creates authentic, entertainment driven customer experiences. Viewers can see products in action, receive personalized recommendations, and build trust through live interaction and social proof. Supported by rapid advancements in AI, AR/VR, and 5G, the live streaming e-commerce market is projected to grow significantly, further transforming the way consumers discover and purchase products (BigCommerce 2025; Softjourn 2025).

### Benefits of Paid Online Advertising

Paid online advertising allows businesses to reach specific demographic and geographic segments with exceptional precision. This not only boosts brand awareness but also increases engagement and website traffic. Because paid ads provide measurable, real time data, businesses can monitor performance, adjust budgets, and optimize return on investment with ease. Additionally, according to (Agency 2025; Impact Group 2020; Oviond 2024), paid advertising offers strategic competitive advantages such as outranking competitors on search engines and social media platforms, which helps strengthen online presence and accelerate customer acquisition.

### Advantages of Digital Content Production

Digital content production enhances brand loyalty and nurtures long term customer relationships. High quality content whether videos, graphics, blogs, or social media posts acts as a bridge between brands and their audiences. It supports lead generation efforts by delivering value, building trust, and encouraging customers to act through well-placed calls to action. Over time, consistent content production significantly increases audience engagement and improves overall marketing impact (Contentoo 2024; Verbolabs 2025)

This integration of paid advertising and digital content production, supported by emerging technologies and personalization trends, positions businesses to succeed in the competitive digital marketplace.

$$\text{Selling price} = \text{Rm}150$$

The total revenue is calculated as:

$$\begin{aligned}\text{Total Revenue} &= 37,500 \times \text{RM}150 \\ &= \text{RM}5,625,000\end{aligned}$$

Since our business is a start-up, we need to invest more heavily in marketing to increase brand awareness and make our products known to a wider audience. Most commerce businesses typically allocate 7–12% of their total revenue to marketing, although that number can go as high as 20% for newer or more growth-oriented brands (Hostinger 2023). Therefore, by focusing on building strong brand awareness and customer acquisition, a marketing budget of 12% of revenue was set, as it strikes a balance between aggressive growth and financial prudence.

$$\text{Marketing Expense} = \text{RM}5,625,000 \times 0.12$$

= RM 675,000

### 9.1.2 Production Expenses

#### Machines:

|                                                                                                                         |                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                                        |  |
| Ra Automatic Soldering Equipment for PCB Assembly Production Line (Dongguan Rongan Electromechanical Equipment Co 2025) | VLT--ST Vacuum Leak Tester (Standard Model) (ExportersIndia 2025)                  |
| RM 17,425.80                                                                                                            | RM 18,500                                                                          |
| Depreciation per year                                                                                                   | Depreciation per year                                                              |
| RM 3485.16                                                                                                              | Rm 3700                                                                            |

As the VLT—ST Vacuum Leak Tester did not list a price, as it requires an inquires for the price. Therefore according to (ExportersIndia 2025) there is a listing for AT2E VLT-ST model which cost aaround 4000 – 6000 USD, thus based on that we can assume the machine is between RM 18,500 – RM 27,800. Therefore, we assume the machine to be RM 18,500.

Furthermore, the machines useful life is set at 5 years because industrial machinery generally has physical durability, slower technological obsolescence, and is classified under IAS 16 as fixed assets with typical economic lifespan 5 – 10 years. Meanwhile, electronic test equipment is assigned a 3-year useful life because electronic components degrade faster, require annual calibration, and face rapid technological updates, consistent with ISO 17025 and common industry practice for precision test devices.

#### Equipment:

As the Go/No-Go Gauges for cap and thread inspection typically ranges between RM 208 to RM 500, based on pricing from metrology suppliers such as Vermont Gage, Meyer Gage, and industrial distributors like MSC Industrial Supply. Furthermore, based on AT2E’s closure gauges generally

fall within RM 400 to RM 900, therefore thorough authorized distributors, we selected RM 500 as a reasonable mid-range estimate consistent with industry benchmark.

|                                                                                   |                                                                                   |                                                                                    |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |
| Go/No-Go Gauges (Custom-made)                                                     | Bore Scope (Shopee 2025g)                                                         | Digital Calliper (Shopee 2025a)                                                    |
| RM 500                                                                            | RM 108.90                                                                         | RM 13.28                                                                           |
| Depreciation per year                                                             | Depreciation per year                                                             | Depreciation per year                                                              |
| RM 166.67                                                                         | RM 36.30                                                                          | RM 4.43                                                                            |

|                                                                                    |                                                                                    |                                                                                     |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |
| Digital Multimeter (Shopee 2025c)                                                  | USB power tester (Shopee 2025b)                                                    | Insulation Resistance tester (Shopee 2025f)                                         |
| RM 46.51                                                                           | RM 11.80                                                                           | RM 348.31                                                                           |
| Depreciation per year                                                              | Depreciation per year                                                              | Depreciation per year                                                               |
| RM 15.50                                                                           | RM 3.93                                                                            | RM 116.10                                                                           |

|                                                                                          |                                                                                   |                                                                                     |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|         |  |  |
| Digital Battery Capacity Checker RC CellMeter 7 2-7S for LiPo LiFe Li-ion (Shopee 2025d) | Torbal 901101 Test Stand (Torbal 2025)                                            | Digital Infrared Thermometer (Shopee 2025e)                                         |
| RM 21.97                                                                                 | Rm 4072.36                                                                        | RM 27.28                                                                            |
| Depreciation per year                                                                    | Depreciation per year                                                             | Depreciation per year                                                               |
| RM 7.32                                                                                  | Rm 1,342.45                                                                       | RM 9.09                                                                             |

*Total Equipment cost = RM5150.11*

## Climate Control Systems

Air-conditioning and ventilation expenses constitute a significant portion of operating costs for Malaysian manufacturing start-ups due to the country's tropical climate and regulatory requirements for workplace air quality. According to (Suruhanjaya Tenaga 2023), cooling systems can account for up to 40–50% of total electricity consumption in commercial and small industrial facilities, making them one of the most energy-intensive components of production environments. This cost intensity is reinforced by national electricity tariff structures, with (Tenaga Nasional 2024) reporting that small commercial consumers typically incur between RM0.43 and RM0.50 per kWh under Tariff C categories. As such, even moderately sized air-conditioning units can represent a recurring operational burden when running for extended production hours.

In estimating asset depreciation for a start-up climate control system, differing useful lifespans are applied to air conditioning units and industrial ventilation equipment due to their distinct operational stresses. Split type or cassette type air conditioning systems are typically depreciated over 5 years, reflecting moderate wear, standardized compressor reliability, and routine servicing that prolongs functional lifespan in commercial settings (Daikin 2023; Malaysian Institute of 2022). In contrast, ventilation systems, including exhaust fans and inline blowers, are generally assigned a 3 years useful life, due to continuous duty operation, exposure to airborne particulates, and accelerated motor or bearing degradation (Malaysia 2023; Panasonic Ventilation 2022). These

assumptions align with common Malaysian SME practices, ensuring depreciation reflects realistic replacement cycles and maintenance demands.

|                                                                                       |                                                                                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|      |            |
| <p>DAIKIN FCC-A R32 Ceiling Cassette<br/>Aircond Non-inverter 2HP Air Conditioner</p> | <p>KDK 17CUH Ceiling Mount Ventilating Fan<br/>Steel Type Ceiling Exhaust Fan Steel Type</p> |
| <p>RM3,214</p>                                                                        | <p>RM479</p>                                                                                 |
| <p>Depreciation per year</p>                                                          | <p>Depreciation per year</p>                                                                 |
| <p>RM 642.80</p>                                                                      | <p>RM159.67</p>                                                                              |

To estimate the annual maintenance cost, according to (AccountingTools 2025), we adopt the widely accepted managerial accounting guideline of 5% of the machine acquisition cost per year. This rate reflects typical wear and tear and routine servicing for production equipment, ensuring our financial model captures a realistic overhead expense.

$$\text{Annual Maintainence Rate} = \text{Machine cost} \times \text{Maintainence Rate}$$

For Machine 1:

$$\begin{aligned}\text{Annual Maintainence Rate} &= 17,425.80 \times 0.05 \\ &= \text{RM } 871.29\end{aligned}$$

For Machine 2:

$$\begin{aligned}\text{Annual Maintainence Rate} &= (18,500 \times 2) \times 0.05 \\ &= \text{RM } 1850\end{aligned}$$

For Climate control systems:

Although both ventilation and air conditioning contain a 5-year warranty, climate control systems require routine maintenance to ensure reliable operation and prevent minor faults. In Malaysia, standard preventative maintenance for air-conditioning and ventilation units typically costs between RM 80 to 600 annually, depending on service scope and usage intensity (AccoolnCare 2023). Life cycle studies indicate that annual maintenance generally ranges from 1% to 10% of

the equipment's replacement value, reflecting wear, environmental exposure, and operational hours (Kauko & Keppo 2021). Accordingly, we adopt a maintenance rate of 1% per year of equipment value during the warranty period and 3% per year thereafter, providing a realistic estimate for budgeting purposes.

During warranty (Air conditioning):

$$\text{Maintenance cost: } (RM3,214 \times 6) \times 0.01 = RM192.84$$

After warranty (Air conditioning):

$$\text{Maintenance cost: } (RM3,214 \times 6) \times 0.03 = RM578.52$$

During warranty (Ventilation):

$$\text{Maintenance cost: } (RM479 \times 4) \times 0.01 = RM19.16$$

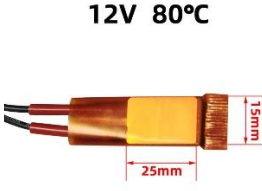
After warranty (Ventilation):

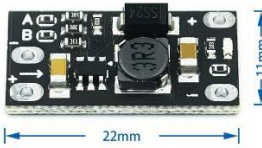
$$\text{Maintenance cost: } (RM479 \times 4) \times 0.03 = RM192.84$$

Total Repair Maintenance cost:

$$RM\ 871.29 + RM\ 1850 + RM192.84 + RM19.16 = RM2933.29$$

### 9.1.3 Purchasing Costs

|                                                                                     |                                                                                     |                                                                                                                                    |                                                                                                                                         |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|  |  |                                                |                                                    |
| ESP 32(Cytron 2025)                                                                 | DS18B20 waterproof digital probe (Shopee 2025n)                                     | PTC Heating plate<br>AC DC 12V<br>Thermostatic<br>Ceramic Heating<br>Plates sheet constant<br>temperature heater<br>(Shopee 2025l) | Pcbfun 1/3/5pcs<br>DC3V-30V Touch<br>Type Single Key<br>Bistable Touch<br>Switch Sensing<br>Module Supports<br>Jog/Latch (Shopee 2025k) |
| RM23.90                                                                             | RM 3.62                                                                             | RM 8.24                                                                                                                            | RM 9.44                                                                                                                                 |

|                                                                                                    |                                                                                                                   |                                                                                                                                             |                                                                                     |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                   |                                  |                                                           |  |
| 3.7V 5000mAh<br>6060100<br>Rechargeable LiPo<br>Li-Po Battery MP3<br>Car GPS DIY<br>(Shopee 2025i) | Mini-Booster Module<br>3.7V to 12V Support<br>5V/8V/9V/12V Output<br>Lithium Battery<br>Booster (Shopee<br>2025j) | DIYMORE 2.25-inch<br>TFT LCD Screen<br>Module St7789 Small<br>Screen 76 * 284 Module<br>Lcd Full View Spi<br>Color Screen (Shopee<br>2025h) | SUS-304<br>aluminium                                                                |
| RM 24.90                                                                                           | RM 0.70                                                                                                           | RM 14.35                                                                                                                                    | RM 22.80                                                                            |

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |
| Double Layer Bubble Wrap (Shopee 2025o)                                             | Box Packaging                                                                        |
| RM 4.80                                                                             | RM 6                                                                                 |

In determining the material cost for the stainless steel (SUS 304) required for the flask, we decided to use RM 20 per kilogram as a reasonable estimate. This figure is based on the current market pricing of SUS 304 in Malaysia. According to (Soon Hoe 2025), a supplier of stainless steel sheets,

a 0.8 mm thick 304 stainless sheet weighs 19.02 kg for a certain sheet area, which aligns with typical industry rates for raw material supply in the country. Additionally, (Steel Industries 2025) provides a catalogue listing 304 stainless steel sheet pricing across various thicknesses, confirming that RM 20 per kg is within the average local market range. Furthermore, global suppliers such as (Alibaba 2025) also list 304 stainless steel at comparable prices, reinforcing the validity of this estimate. By using RM 20 per kilogram, I am able to establish a baseline material cost that is realistic, market-driven, and supported by multiple references, which ensures that the subsequent calculations for manufacturing costs are grounded in current industry standards. Therefore, with founded design of 1.14kg per unit:

$$\begin{aligned} \text{Bare Bone case} &= 1.14\text{kg} \times \text{RM } 20 \\ &= \text{RM}22.80 \end{aligned}$$

For our smart 2 L vacuum flask, we decided to use premium gift packaging that is designed in house to ensure both protection and presentation of the product. The packaging is tailored to accommodate not only the flask but also its charging wire, socket, and accessories, with a custom moulded EVA foam or thermoformed insert that secures each component in place. This design choice enhances the unboxing experience, which is critical for technology driven and gift-oriented products, while also providing sufficient protection during transport.

According to (Alibaba.com 2025), the estimated cost of such premium packaging is between RM 10 to RM 25 per unit, based on references from packaging suppliers in Malaysia and Alibaba. For example, rigid gift boxes and thermoformed inserts for electronics typically range from RM 6–15 per unit, with additional costs for printed manuals, warranty cards, and protective shipping layers bringing the total to approximately RM 10–25 per set. These costs align with industry practices where packaging can account for 3–15% of the product cost and are consistent with premium electronic or consumer goods packaging standards (Alibaba.com 2025; Shopee 2025m);(Solutions 2025). By designing the packaging ourselves, we are able to optimize the layout, branding, and protection while controlling costs within this range. Therefore, the packaging is decided of RM 6, meanwhile having the bubble wrap it will add an additional of RM 3.80 to properly secure and protect the flask, charging wire, socket, and accessories inside the box.

$$\text{Total Component Cost for per unit} = \text{Rm } 118.75$$

Furthermore, to estimate the logistics cost for delivering components, we assumed an average weight of 0.5 kg per unit, based on typical consumer smart vacuum flask components such as the flask body, lid, and small electronic parts (Tan, Lim & Lee 2021). This weight assumption allows us to calculate shipping costs per order and per unit, which are included as part of the variable costs in our production model. The delivery cost per unit is then determined by dividing the total shipping cost of each order by the number of units per shipment. Although the exact component weight may vary slightly, using a conservative estimate ensures that our cost model does not underestimate the financial burden of logistics.

$$\text{Units per order} = 562$$

$$\text{Weights per unit } 0.5 \times 562 = 281\text{kg/shipment}$$

$$\text{Shipping rate} = \text{RM}0.50/\text{kg}$$

$$\begin{aligned}\text{Shipping cost per order} &= 0.50 \times 281 \\ &= \text{Rm } 140.50\end{aligned}$$

$$\begin{aligned}\text{Delivery cost per unit} &= \frac{140.50}{562} \\ &\approx \text{RM}0.25\end{aligned}$$

$$\begin{aligned}\text{Annual Delivery cost} &= \text{RM } 0.25 \times 37500 \\ &= \text{RM}9,375\end{aligned}$$

#### 9.1.4 Personnel Expenses

As the workers work 6 days a week and by excluding the holidays in Malaysia, we have a total of 300 working days.

Therefore, to calculate the units produce per day:

$$\begin{aligned}\text{Units per day} &= \frac{37,500}{300} \\ &= 125 \text{ units/day}\end{aligned}$$

Labour time in each section:

Table 14: Task Description of each Section and their Estimated Time

| Section | Task Description                             | Estimated Time (mins/unit) |
|---------|----------------------------------------------|----------------------------|
| 1       | Inspect bare flask components                | 2                          |
| 2       | Electronics assembly / soldering (automated) | 1                          |
| 3       | Mechanical assembly                          | 8                          |
| 4       | Electronic testing                           | 3                          |
| 5       | Leak testing                                 | 3                          |
| 6       | Safety & performance testing                 | 4                          |
| 7       | QC / visual testing                          | 2                          |
| 8       | Packing and labelling                        | 2                          |

$$\text{Total Labour time per unit} = \frac{2 + 1 + 8 + 3 + 3 + 4 + 2 + 2}{60}$$

$$= 0.42 \text{ hours/unit}$$

$$\begin{aligned} \text{Total Labour hours per day} &= 125 \times 0.42 \\ &= 52 \text{ hours/day} \end{aligned}$$

As we assume each shift is 12 hours, with 1 shift:

$$\begin{aligned} \text{Workers per shift} &= \frac{\text{Total labour hours/day}}{\text{Hours per shift} \times \text{Number of shifts}} \\ &= \frac{52}{12 \times 1} \\ &= 4.33 \\ &\approx 5 \text{ workers} \end{aligned}$$

According to (PLT 2024), in Malaysia the minimum wages of salaries for a labour worker is RM 1700 per month

$$\begin{aligned} \text{Total Wages in year} &= 5 \times 1700 \times 12 \\ &= \text{RM}102,200 \end{aligned}$$

### Additional Personnel:

For the proposed start-up, our project employs six non-production personnel: A Production Supervisor, Accountant, Administrative Clerk, Maintenance Technician, Sales Coordinator, and Marketing Executive. Based on latest market data, the average monthly wages for these roles in Malaysia range from RM 1,956 (Administrative Clerk) to RM 4,800 (Marketing Executive). Factoring in mandatory employer statutory contributions including EPF at 13%, SOCSO at 1.75%, and EIS at 0.2% the total annual employment cost per role increases by approximately 15–20% (*SOCSSO, EPF and EIS Employer Contribution Guide* 2023).

*Table 15: Salary for Workers*

| <b>Role</b>           | <b>No. of employees</b> | <b>Monthly Salary (MYR)</b>                            | <b>Annual Salary (MYR)</b> | <b>Monthly Contributions (EPF+SOCSSO+EIS)</b> | <b>Total Annually Salary + Contributions (MYR)</b> |
|-----------------------|-------------------------|--------------------------------------------------------|----------------------------|-----------------------------------------------|----------------------------------------------------|
| Production Supervisor | 1                       | 3,313('Production Supervisor Salary in Malaysia' 2025) | 39,756                     | 6,154                                         | 45,910                                             |

|                        |   |                                                                            |        |       |         |
|------------------------|---|----------------------------------------------------------------------------|--------|-------|---------|
| Accountant             | 1 | 4,389('Accountant Salary in Malaysia' 2025)                                | 52,668 | 7,954 | 60,622  |
| Administrative Task    | 1 | 1,956('Administrative Clerk Salary in Malaysia' 2025)                      | 23,472 | 3,549 | 27,021  |
| Maintenance Technician | 1 | 3,000( <i>Salary Guide for Sales and Marketing Roles in Malaysia</i> 2023) | 36,000 | 5,580 | 41,580  |
| Sales Coordinator      | 1 | 3,500( <i>Salary Guide for Sales and Marketing Roles in Malaysia</i> 2023) | 42,000 | 6,516 | 48,516  |
| Marketing Executive    | 1 | 4,800( <i>Salary Guide for Sales and Marketing Roles in Malaysia</i> 2023) | 57,600 | 8,928 | 66,528  |
| Total cost             |   |                                                                            |        |       | 290,177 |

### 9.1.5 Financial and administrative expenses

#### Insurance

To safeguard our production assets and mitigate operational risks, the company will obtain essential machinery and liability insurance appropriate for a manufacturing start-up. According to Malaysian SME insurer guidelines, machinery breakdown insurance premiums generally fall between 0.3% and 1.0% of the equipment value annually ('Engineering & Machinery Breakdown Insurance Guide' 2022; 'Machinery Breakdown Insurance' 2023). Fire and property insurance for industrial equipment typically ranges from 0.15% to 0.25% of insured asset value ('Fire & Industrial Property Insurance Product Disclosure Sheet' 2023). Furthermore, basic public liability coverage for small enterprises in Malaysia is reported to cost approximately RM 200 to RM 400 per year, depending on business size and risk exposure ('SME Business Shield: Liability & Asset Protection Overview' 2022).

Given the higher operational uncertainty associated with new manufacturing ventures, this study adopts a conservative combined machinery insurance rate of 1% per annum, consistent with the upper bound of Malaysian SME insurance benchmarks. Our key production equipment includes

an automated soldering machine valued at RM 17,425.80 and a VLT-ST Vacuum Leak Tester valued at RM 18,500, resulting in a total insured asset value of RM 35,925.80. Applying the selected 1% insurance rate yields an annual machinery insurance expense of:

$$\begin{aligned}\text{Annual insurance cost} &= 0.01 \times \text{RM } 35,925.80 \\ &= \text{RM } 359.26\end{aligned}$$

Using a higher-end rate ensures the estimated insurance cost sufficiently covers risks related to equipment malfunction, downtime, accidental damage, and potential repair or replacement needs. This approach provides a realistic cost structure while remaining aligned with Malaysian industry insurance standards for start-ups.

### Interest paid

To finance part of the start-up's initial investment, the company plans to secure a bank loan. In Malaysia, small and medium-sized enterprises (SMEs) typically obtain loans at interest rates ranging from 4% to 7% per annum, depending on the lender, loan scheme, and the borrower's credit profile (*Guidelines on SME Loans and Financing* 2023; *SME Financing Schemes* 2023). Government supported SME loan schemes often provide slightly lower interest rates, approximately 4%–5%, whereas commercial bank loans for new businesses generally range from 5%–6% annually.

For our start-up, we assume a conservative interest rate of 5% per annum on a loan of RM 250,000 to partially fund initial machinery, equipment, and inventory purchases. This results in an estimated annual interest expense of RM 12,500 or approximately RM 1,042 per month, calculated as:

$$\begin{aligned}\text{Interest Expense} &= \text{Loan Amount} \times \text{Interest Rate} \\ &= 250,000 \times 0.05 \\ &= \text{RM}12,500\end{aligned}$$

This assumption provides a realistic estimate of financing costs while ensuring that our financial projections account for the burden of debt service. The interest expense will be categorized under Financial and Administrative Expenses in the projected income statement, reflecting the cost of capital used to support operational and investment activities.

### Licensing fees

Table 16: Licensing Fees

| License / Certification                             | Estimated Cost (RM)                                           |
|-----------------------------------------------------|---------------------------------------------------------------|
| SSM (Companies Commission of Malaysia) Registration | 1,200 ( <i>Company Registration Fees and Procedures</i> 2023) |
| Business Premise License                            | 400 ( <i>Business Premise Licensing Guidelines</i> 2023)      |

|                                                                 |                                                                                           |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Regulatory Compliance (MOH / Food-Grade Material Certification) | 800 ( <i>Regulatory Compliance for Food Contact Materials 2023</i> )                      |
| SIRIM Certification                                             | 2500 ( <i>Product Certification and Testing Services for Consumer Electronics 2023</i> )  |
| Electrical / EMC Certification                                  | 1,500 ( <i>Product Certification and Testing Services for Consumer Electronics 2023</i> ) |
| Trademark Registration                                          | 1,000 ( <i>Trademark Registration Guide and Fees 2023</i> )                               |
| <b>Total Cost</b>                                               | <b>RM 7400</b>                                                                            |

### 9.1.6 Fixed costs or Variable costs

Table 17: Fixed Costs and Variable Costs

| Fixed Costs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Variable Costs                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Machinery Purchase</li> <li>• Equipment Purchase</li> <li>• SSM Company Registration</li> <li>• Business Premise License</li> <li>• Regulatory Compliance / Food-grade Certification</li> <li>• SIRIM Certification</li> <li>• Electrical / EMC Certification</li> <li>• Trademark Registration</li> <li>• Insurance (Machinery + Liability)</li> <li>• Maintenance &amp; Repairs</li> <li>• Bank Interest on Loan</li> <li>• Marketing Expenses</li> <li>• Climate control system</li> </ul> | <ul style="list-style-type: none"> <li>• Inventory / Raw Materials</li> <li>• Logistics / Delivery</li> <li>• Labour (Production)</li> <li>• Utilities (Electricity for machines)</li> </ul> |

### 9.1.7 Required sales to cover both fixed and variable costs

$$\begin{aligned}
 \text{Contribution per unit} &= \text{Selling Price per Unit} - \text{Variable Cost per Unit} \\
 &= \text{RM } 150 - \text{RM } 118.75 - \text{RM } 11.44 - \text{RM } 0.25 - \text{RM } 0.081 \\
 &= \text{RM } 19.48
 \end{aligned}$$

*Total Fixed Cost*

$$\begin{aligned}
 &= 17,425.80 + (18500 \times 2) + 5150.11 + 7400 + 359.26 + 1796.29 + 12500 + 675,000 \\
 &\quad + (\text{RM } 3,214 \times 6) + (\text{RM } 479 \times 4)
 \end{aligned}$$

$$= RM777,831.46$$

$$\begin{aligned} \text{Break even Units} &= \frac{777,831.46}{19.48} \\ &= 39,929.7 \\ &\approx 39,930 \text{ units} \end{aligned}$$

Bulk purchasing is a widely adopted strategy in manufacturing to reduce per unit costs and improve operational efficiency. Economies of scale allow firms to negotiate lower prices for components when ordering larger quantities, typically resulting in discounts ranging from 5% to 20% depending on the supplier and annual order volume (Heizer & Render 2020).

Therefore, the new component cost can be lowered and based on this, we assume to have a discount of 15%.

$$RM118.75 \times 0.15 = RM17.8125$$

$$\text{Total new component cost} = RM100.94$$

$$\begin{aligned} \text{New Contribution per unit} &= \text{Selling Price per Unit} - \text{Variable Cost per Unit} \\ &= RM 150 - RM100.94 - Rm11.44 - RM0.25 - RM0.081 \\ &= RM37.289 \\ &\approx RM37.29 \end{aligned}$$

$$\begin{aligned} \text{New Break even Units} &= \frac{777,831.46}{37.29} \\ &= 20,858.98 \\ &\approx 20,859 \text{ units} \end{aligned}$$

## 9.2 Establishment Costs

The establishment cost for this business plan is a one-time expense that need to be incurred to start the business plan. Establishment costs refer to all one-time expenditures required to set up ThermaCorex before operations and sales begin. These costs include facility preparation, equipment, branding, regulatory compliance and initial inventory purchases necessary to launch the smart vacuum flask business.

ThermaCorex will be based in a light industrial factory, which combines a small-scale assembly line, inventory storage and administrative offices within a single facility. Such integrated spaces are commonly adopted by hardware and IoT start-ups due to their operational flexibility, streamlined material flow and cost efficiency. Concentrating all core activities in one location enables faster assembly and quality assurance processes, minimises travel distance between production, warehousing and administrative functions, and enhances the accuracy of inventory handling. This co-located arrangement also facilitates closer interaction between R&D, operations

and management teams, thereby supporting more effective decision-making and coordination. Furthermore, this type of factory provides the capacity for future expansion, allowing ThermaCorex to scale its production output or incorporate additional assembly modules as market demand increases.

*Total Establishment cost*

$$= \text{Equipment Cost} + \text{Machine Cost} + \text{Licensing} + \text{Initial Inventory} \\ + \text{Marketing Expenses} + \text{Labour expenses} + \text{Climate control system}$$

$$= 5150.11 + 17,425.80 + (18,500 \times 2) + 7,400 + 50,000 + 675,000 + (RM3,214 \times 6) \\ + (RM479 \times 4)$$

$$= RM813,175.91$$

### 9.3 Running Costs

According to (nVentic 2023), to estimate the holding cost for our inventory, we adopted a holding cost rate of 20% of the unit cost per year. This rate is commonly used for start-ups and small businesses, as holding costs typically range between 10% and 25% of the inventory value, depending on storage requirements, capital cost, insurance, and risk of product obsolescence. Since our product is a consumer electronic item with components that may depreciate or become outdated over time, a higher holding cost rate provides a more realistic and conservative estimate. Selecting 20% places us at the upper end of the standard industry range, ensuring we capture expenses such as warehouse storage, utilities, inventory handling, and the opportunity cost of tied-up capital. This assumption helps create a more accurate cost model and reduces the risk of underestimating the true financial burden of maintaining inventory.

$$\text{Demand, } D = 37,500 \text{ units/year}$$

$$\text{Holding rate, } H_{\text{rate}} = 0.2 \times 118.75 \\ = RM23.75 \text{ per unit/year}$$

Furthermore, it is assumed an ordering cost of RM 100 per order, based on an EOQ-model from (Ahmad et al. 2021). This value reflects our operational scale and the administrative effort involved in managing a relatively high annual demand of 37,500 units. Although the typical ordering cost range for businesses is RM 50–RM 200 per order, we chose RM 100 to account for bulk ordering, supplier coordination, documentation, and procurement management. Despite streamlined digital processes that reduce administrative effort, the larger volume of units increases the complexity and resources required per order.

This value is consistent with industry references, which indicate that ordering costs vary based on paperwork, labour, shipping coordination, and supplier communication, especially for businesses handling larger volumes. Using RM 100 per order ensures that our EOQ model accurately captures

the real cost of placing orders while remaining realistic and aligned with standard practices for start-ups operating at medium to large scale.

$$\begin{aligned} \text{Holding Cost} &= \frac{Q^*}{2} H \\ &= \frac{562}{2} (23.75) \\ &= \text{RM}6,673 \end{aligned}$$

As shown above the annual setup cost can be calculated:

$$\begin{aligned} \text{Annual setup cost} &= \frac{D}{Q} S \\ \text{Annual Setup cost} &= \frac{37,500}{562} (100) \\ &= 6,672.598 \\ &\approx \text{RM } 6,673 \end{aligned}$$

$$\begin{aligned} \text{Total Annual Cost, } TC &= \text{Setup Cost} + \text{Holding Cost} \\ &= \text{RM}6,673 + \text{RM}6,673 \\ &= \text{RM } 13,346 \end{aligned}$$

## Utilities Cost

Electricity cost is an essential component of estimating manufacturing overhead. According to the latest tariff updates released by Malaysia SME News Portal, the average base electricity tariff for Malaysia is 45.40 sen/kWh (RM 0.454/kWh) under the streamlined structure introduced by Tenaga Nasional Berhad (TNB) ('New electricity structure to benefit 23.6 million users with fairer, more transparent rates'). A similar update by ('July 2025 TNB Tariff Update: What Malaysian businesses need to know') reports that low voltage general non-domestic users typical for small industrial start-ups face an effective tariff of RM 0.2703/kWh, reflecting Malaysia's efforts to maintain competitive energy costs for businesses. Considering both published sources and the need for a conservative estimate suitable for early-stage manufacturing planning, this study adopts an electricity cost assumption of RM 0.45/kWh. This figure appropriately reflects current tariff levels while providing a prudent buffer for potential fluctuations.

## Machine 1:

Ra Automatic Soldering Equipment for PCB Assembly Production Line

$$\text{Energy (kWh)} = kW \times \text{Time per Unit (h)}$$

$$0.15kW/unit \times 0.12h = 0.018kWh/unit$$

The Electricity cost is RM 0.45 per kWh:

$$\begin{aligned} \text{Electricity cost/unit} &= 0.018 \times 0.45 \\ &= RM0.081/unit \end{aligned}$$

$$\begin{aligned} \text{Annual Electricity cost (machine 1)} &= RM0.081 \times 37500 \\ &= RM 3037.50 \end{aligned}$$

### Machine 2:

VLT--ST Vacuum Leak Tester (Standard Model)

$$\text{Energy (kWh)} = kW \times \text{Time per Unit (h)}$$

$$0.15kW/unit \times 0.12h = 0.018kWh/unit$$

The Electricity cost is RM 0.45 per kWh:

$$\begin{aligned} \text{Electricity cost/unit} &= 0.018 \times 0.45 \\ &= RM0.081/unit \end{aligned}$$

$$\begin{aligned} \text{Annual Electricity cost (machine 2)} &= RM0.081 \times 37500 \times 2 \\ &= RM6075 \end{aligned}$$

### Climate Control System:

Air conditioning

$$\text{Energy (kWh)} = kW \times \text{Time (h)}$$

$$1.68kW \times 12h = 20.16kW/h$$

The Electricity cost is RM 0.45 per kWh:

$$\begin{aligned} \text{Daily Electricity Cost} &= 20.16 \times 0.45 \\ &= RM9.072/unit \end{aligned}$$

$$\begin{aligned} \text{Annual Electricity cost} &= RM9.072 \times 300 \times 6 \\ &= RM 16,329.60 \end{aligned}$$

Ventilation

$$\text{Energy (kWh)} = kW \times \text{Time per Unit (h)}$$

$$1.68kW \times 12h = 20.16kW/h$$

The Electricity cost is RM 0.45 per kWh:

$$\begin{aligned}\text{Daily Electricity Cost} &= 20.16 \times 0.45 \\ &= \text{RM}9.072/\text{unit}\end{aligned}$$

$$\begin{aligned}\text{Annual Electricity cost} &= \text{RM}9.072 \times 300 \times 4 \\ &= \text{RM } 10,886.40\end{aligned}$$

$$\begin{aligned}\text{Total Annual Electricity cost} &= \text{RM}3037.50 + \text{RM}6075 + \text{RM } 16329.60 + \text{RM } 10886.40 \\ &= \text{RM}36,328.50\end{aligned}$$

### Water Bill

The table below shows the domestic water rate for Penang region from Perbadanan Bekalan Air Pulau Pinang Sdn Bhd(PBA).

Table 18: Rates for Domestic Water Supply Services in Penang

| <b>Rates for Domestic Water Supply Services</b> |                              |
|-------------------------------------------------|------------------------------|
| Amount of Consumption                           | Rate per $m^3$ (1000 litres) |
| 0-20 $m^3$                                      | RM0.62 per $m^3$             |
| 20 $m^3$ -35 $m^3$                              | RM1.17 per $m^3$             |
| Exceeding 35 $m^3$                              | RM2.07 per $m^3$             |
| <b>Minimum charge</b>                           | <b>RM6.20 per month</b>      |

A realistic assumption for workplace use is about 40–60 litres per worker per day. By taking the median value of 50 litres and assuming there are 5 workers in a factory working 22 days per month, the monthly water consumption is 5500 litres. The calculation for monthly water bill is as follows:

### Monthly Water Bill Calculation

$$\text{Amount of consumption} = 5.5 \text{ } m^3 \text{ per month}$$

$$\begin{aligned}\text{Actual Charge for Water Bill} &= 5.5 \times \text{RM}0.62 \\ &= \text{RM}3.41\end{aligned}$$

Since the actual charge for the water bill is below the minimum charge, a flat fee of RM6.20 is charged instead.

### Annual Water Bill Calculation

$$\begin{aligned}\text{Annual Water Bill} &= 12 \times \text{RM}6.20 \\ &= \text{RM}74.40\end{aligned}$$

$$\begin{aligned}\text{Total Utility cost} &= \text{RM}36,328.50 + \text{RM}74.40 \\ &= \text{RM}36,402.90\end{aligned}$$

## Rental

Industrial property rental rates in Bayan Lepas, Penang, remain firmly in the range of RM 2.00 to 3.60 psf per month, based on recent market analyses (Cbre & Wtw 2024). This aligns closely with Penang Development Corporation's published rates for semi-detached factory units within the Bayan Lepas Industrial Park, which range from RM 2.66 to RM 2.93 psf/month (Penang Development 2025). Given these benchmarks, assuming a factory rental rate of RM 2.90 psf/month for a start-up production facility is consistent with prevailing market conditions and supports a realistic forecast of long-term operating expenditure.

$$\text{Assuming the production area} = 5000\text{ft}^2$$

$$\begin{aligned}\text{By using RM2.90 psf, monthly rent} &= 5000 \times 2.90 \\ &= \text{RM}14,500\end{aligned}$$

$$\begin{aligned}\text{Annual rent} &= \text{RM } 14,500 \times 12 \\ &= \text{RM } 174,000\end{aligned}$$

## 9.4 Sales Forecast

Forecasting future demand is a critical step in determining ThermaCorex's production capacity, inventory requirements and overall financial planning. As a newly introduced product with no historical sales data, ThermaCorex relies on qualitative forecasting methods particularly market surveys to estimate its initial Year 1 demand. Malaysia's consumer electronics market is undergoing rapid transformation driven by widespread wireless connectivity, expanding IoT adoption, increased R&D activity, and the growing popularity of wearable and smart lifestyle devices. According to the Malaysian Investment Development Authority (MIDA), these technological shifts present strong opportunities for innovative consumer-tech products to scale within the country. Leveraging these market trends, ThermaCorex's early-stage demand forecast aims to reflect the realistic purchasing behaviour of Malaysian consumers in a technology-driven environment.

According to GlobalData, Malaysia's e-commerce industry is projected to grow at a compound annual growth rate (CAGR) of 8.5% between 2024 and 2028, reaching an estimated market value of MYR 67.1 billion by 2028. A CAGR of 8.5% indicates that the sector is expected to expand by 8.5% every year on a compounding basis. This means that if the market grows at this rate annually, its value increases by multiplying the previous year's figure by 1.085 each year. This steady annual increase highlights the strong momentum of Malaysia's digital commerce ecosystem and supports a positive outlook for consumer-tech products sold through online channels.

### First Year Sale

$$\begin{aligned}
 \text{Estimated sales} &= \text{Selling price per unit} \times \text{Sales Volume} \\
 &= \text{RM150} \times 37500 \\
 &= \text{RM5625000}
 \end{aligned}$$

According to Statista website, Malaysia inflation rate as of now is 1.59% this year while it is forecasted to be 2.2% in 2026 and 2% in 2027. Hence selling price per unit must be adjusted also according to the inflation rate.

Table 19: Price per unit in 2025, 2026 and 2027

| Product            | Price per Unit in 2025 | Price per Unit in 2026 (2.2% inflation compared to previous year) | Price per Unit in 2027 (2% inflation compared to previous year) |
|--------------------|------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------|
| Smart Vacuum Flask | RM150                  | RM153.30                                                          | RM156.37                                                        |

Table 20: Annual Production, Demand Increase, Inflation and Inventory for 2025, 2026 and 2027

| Year                      | 2025    | 2026    | 2027    |
|---------------------------|---------|---------|---------|
| Annual Production (units) | 37500   | 40688   | 44146   |
| Demand Increase %         | 8.5     | 8.5     | 8.5     |
| Inflation %               | 0       | 2.2     | 2       |
| Inventory (RM)            | 5625000 | 6898500 | 8262000 |

## 9.5 Income Statement & Profit Forecast

In addition to revenue generated from the core ThermaCorex smart vacuum flask, the business will also benefit from accessories sales, which serve as an important supplementary income stream. Accessories such as replacement lids, silicone sleeves, thermal protective covers, cleaning brushes, and customised carry straps enhance the usability and longevity of the product, making them attractive add-ons for customers. These items typically carry higher profit margins than the main product due to lower production and material costs, allowing the company to maximise earnings from each customer. Furthermore, accessories naturally encourage repeat purchases, as users may replace worn-out components or upgrade to new designs over time. By offering a well-curated range of complementary accessories, ThermaCorex can increase the average revenue per customer

while strengthening brand loyalty and providing a steady, recurring source of income beyond the initial flask purchase.

| ThermaCorex Sdn. Bhd.<br>Profit & Loss 2025- 2027 |  |  |  |                   |                   |                   |
|---------------------------------------------------|--|--|--|-------------------|-------------------|-------------------|
| Year                                              |  |  |  | 2025              | 2026              | 2027              |
|                                                   |  |  |  | \$                | \$                | \$                |
| Sales Revenue                                     |  |  |  | RM 5,625,000.00   | RM 6,237,470.40   | RM 6,754,338.00   |
| Cost of Goods Sold                                |  |  |  | RM (3,785,250.00) | RM (4,107,046.72) | RM (4,456,097.24) |
| <b>Gross Profit</b>                               |  |  |  | RM 1,839,750.00   | RM 2,130,423.68   | RM 2,298,240.76   |
| <i>Operation Expenses</i>                         |  |  |  |                   |                   |                   |
| Marketing                                         |  |  |  | RM 675,000.00     | RM 675,000.00     | RM 675,000.00     |
| Wages                                             |  |  |  | RM 392,377.00     | RM 392,377.00     | RM 392,377.00     |
| Delivery                                          |  |  |  | RM 9,375.00       | RM 9,375.00       | RM 9,375.00       |
| Maintenance                                       |  |  |  | RM 2,933.29       | RM 2,933.29       | RM 2,933.29       |
| Electricity                                       |  |  |  | RM 36,402.90      | RM 36,402.90      | RM 36,402.90      |
| Inventory Setup & Holding                         |  |  |  | RM 13,346.00      | RM 6,951.05       | RM 7,240.37       |
| Insurance                                         |  |  |  | RM 359.26         | RM 359.26         | RM 359.26         |
| Rentals                                           |  |  |  | RM 174,000.00     | RM 174,000.00     | RM 174,000.00     |
| Depreciation (Machinery + Equipment)              |  |  |  | RM 12,586.95      | RM 12,586.95      | RM 12,586.95      |
| <b>Total Operation Expenses</b>                   |  |  |  | RM (1,316,380.40) | RM (1,309,985.45) | RM (1,310,274.77) |
| <b>EBIT (Earnings before Interest &amp; Tax)</b>  |  |  |  | RM 523,369.60     | RM 820,438.23     | RM 987,965.99     |
| <i>Additional Expenses</i>                        |  |  |  |                   |                   |                   |
| <b>Licensing Fee</b>                              |  |  |  | RM (7,400.00)     | RM (7,400.00)     | RM (7,400.00)     |
| <b>Interest Expense</b>                           |  |  |  | RM (12,500.00)    | RM (12,500.00)    | RM (12,500.00)    |
| <b>Net Profit</b>                                 |  |  |  | RM 503,469.60     | RM 800,538.23     | RM 968,065.99     |

## 9.6 Financial Statements

The projected balance sheet for ThermaCorex incorporate the calculation of establishment costs and the determination of required capital, which amounts to RM150,000. To support initial development, technology refinement, and early-stage operations, an investor will contribute RM500,000 to finance product development and market expansion.

Six months after commencing operations, ThermaCorex secures a bank loan of RM250,000 to address underestimated operational and manufacturing costs, particularly related to smart heating components and electronic system integration. An investor return of 7% is projected, amounting to RM35,000 annually, which will be paid from the company's operating profit once cash flow stabilises.

No dividends will be distributed to partners during the projection period (2025~2027), as the business is expected to reinvest earnings into improving product design, expanding production capabilities, and strengthening market penetration. It is also assumed that inventory will be fully

depleted at the end of each reporting period and replenished at the beginning of the following year to maintain accurate cost and stock valuation aligned with annual demand.

| ThermaCorex Sdn. Bhd.                     |  |  |  |                 |                 |                 |
|-------------------------------------------|--|--|--|-----------------|-----------------|-----------------|
| Projected Balance Sheet as of 31 December |  |  |  |                 |                 |                 |
|                                           |  |  |  | 2025            | 2026            | 2027            |
|                                           |  |  |  | \$              | \$              | \$              |
| <i>Current Assets</i>                     |  |  |  |                 |                 |                 |
| Cash                                      |  |  |  | RM 1,328,116.50 | RM 2,101,091.57 | RM 3,041,594.40 |
| Stock of Supplies (Inventory)             |  |  |  | RM 28,364.14    | RM 28,364.14    | RM 28,364.14    |
|                                           |  |  |  | RM 1,356,480.64 | RM 2,129,455.71 | RM 3,069,958.54 |
| <i>Non-Current Assets</i>                 |  |  |  |                 |                 |                 |
| Machinery                                 |  |  |  | RM 54,425.80    | RM 46,988.96    | RM 39,552.12    |
| Equipment                                 |  |  |  | RM 5,150.11     | RM 5,150.11     | RM 5,150.11     |
| Depreciation                              |  |  |  | RM (12,586.95)  | RM (12,586.95)  | RM (12,586.95)  |
| <b>PPE</b>                                |  |  |  | RM 46,988.96    | RM 39,552.12    | RM 32,115.28    |
| <b>Total Assets</b>                       |  |  |  | RM 1,403,469.60 | RM 2,169,007.83 | RM 3,102,073.82 |
| <i>Liabilities</i>                        |  |  |  |                 |                 |                 |
| Bank Loan                                 |  |  |  | RM 250,000.00   | RM 250,000.00   | RM 250,000.00   |
| Investor ROI payout                       |  |  |  | RM 35,000.00    | RM 35,000.00    | RM 35,000.00    |
|                                           |  |  |  | RM 285,000.00   | RM 285,000.00   | RM 285,000.00   |
| <i>Equity</i>                             |  |  |  |                 |                 |                 |
| Investor Capital                          |  |  |  | RM 500,000.00   | RM 500,000.00   | RM 500,000.00   |
| Owner's Capital                           |  |  |  | RM 150,000.00   | RM 150,000.00   | RM 150,000.00   |
| Retained Profit/Loss                      |  |  |  | RM 468,469.60   | RM 1,234,007.83 | RM 2,167,073.82 |
|                                           |  |  |  | RM 1,118,469.60 | RM 1,884,007.83 | RM 2,817,073.82 |
| <b>Total Liability + Owner's Equity</b>   |  |  |  | RM 1,403,469.60 | RM 2,169,007.83 | RM 3,102,073.82 |

## 10.0 Business Future Viability

Based on the financial ratios computed for ThermaCorex over the 3-year period, the business demonstrates strong profitability and improving operational efficiency.

| Year                         | 2025        | 2026        | 2027        |
|------------------------------|-------------|-------------|-------------|
| Gross Profit Margin          | 32.71%      | 34.16%      | 34.03%      |
| Net Profit Margin            | 8.95%       | 12.83%      | 14.33%      |
| Return On Asset              | 35.87%      | 36.91%      | 31.21%      |
| Asset Turnover               | 6.016042781 | 6.671091337 | 7.223890909 |
| Asset Turnover Period (days) | 60.67111111 | 54.7136865  | 50.52678738 |

### 10.1 Profitability

- Return on Total Assets (ROA)

ThermaCorex shows consistently high ROA values, indicating strong capability in utilising its assets to generate earnings. ROA remains stable at around 35~37% in 2025 and 2026, before adjusting slightly to 31.21% in 2027. Despite the minor decline, the figures remain healthy and reflect effective asset deployment.

- Net Profit Margin (NPM)

The business displays strong profitability growth, increasing from 8.95% in 2025 to 14.33% in 2027. This trend indicates that ThermaCorex can control operational expenses more effectively over time while expanding its revenue base.

- Gross Profit Margin (GPM)

The gross profit margin also remains stable and slightly improves from 32.71% in 2025 to around 34% by 2027. This signals efficient production cost management and effective pricing strategies, ensuring sustainable contribution margins over the years.

Overall, the profitability indicators confirm that ThermaCorex is moving towards a more financially secure and stable business model, with strong returns relative to its assets and consistent improvement in margins.

### 10.2 Efficiency

The asset turnover ratio shows continuous improvement, increasing from 6.01 times in 2025 to 7.22 times in 2027. This indicates that each ringgit invested in assets generates more revenue each year, reflecting stronger productivity and better utilisation of operational resources.

Correspondingly, the asset turnover period which measures the number of days needed to convert assets into sales improves significantly from 60.67 days in 2025 to 50.53 days in 2027. This reduction means ThermaCorex is generating revenue faster with each passing year.

These improvements suggest efficient use of key operational assets such as production systems, automation tools, and inventory. However, as the company scales, it will be important to continuously monitor the performance and maintenance needs of high-value equipment to prevent depreciation-related inefficiencies.

## 11.0 Exit Strategy

For ThermaCorex, an ideal exit strategy is acquisition by a larger engineering, HVAC, or smart-building technology company. With the rising demand for energy-efficient thermal management systems and smart automation solutions, ThermaCorex becomes an attractive acquisition target for corporations seeking to expand into advanced thermal control, IoT-integrated systems, or sustainable industrial solutions.

ThermaCorex's core innovation which is the sensor-driven thermal regulation system, embedded automation, and efficient heat-exchange design can position it competitively within the smart-infrastructure and industrial automation market. These capabilities allow real-time temperature optimisation, reduced energy consumption, and reliable performance in both commercial and industrial settings. Larger companies can integrate ThermaCorex's technology to shorten their own R&D cycles, strengthen their sustainability initiatives, and enhance product offerings in smart building management.

Potential acquirers include:

- HVAC and industrial engineering companies such as Daikin, Mitsubishi Electric or Johnson Controls, which continuously seek high-efficiency thermal solutions.
- Technology companies like Siemens or Honeywell, which specialise in smart infrastructure and IoT-enabled building automation.
- Investment or private equity firms focused on sustainable energy solutions and industrial efficiency technologies, which may view ThermaCorex as a high-growth strategic addition to their portfolio.

This exit strategy maximises investor returns by leveraging the company's increasing valuation driven by innovation, performance improvements and strong operational efficiency. It also ensures long-term scalability, as ThermaCorex requires significant capital, advanced facilities and broader market access to fully expand its product line and integrate into large-scale industrial ecosystems. An acquisition provides the platform and resources necessary for ThermaCorex to grow beyond its initial operational scope while maintaining technological leadership in the thermal systems industry.

## 12.0 Conclusion

ThermaCorex can demonstrate a strong potential to meet growing consumer demand for smart, portable, and energy-efficient heating solutions. Its core innovation, the intelligent thermal flask with an internal heating element addresses modern lifestyle needs such as convenience, safety, and consistent temperature control. The financial analysis from 2025 to 2027 indicates a positive outlook, with improvements in gross profit margin, net profit margin, and efficient utilisation of assets. These gains are driven by strategic cost control, targeted marketing, and scalable product engineering.

As consumer interest in smart lifestyle gadgets and personal thermal devices continues to rise, ThermaCorex is well-positioned to build a strong market presence among tech savvy working adults, travellers, students, and outdoor users. The business's commitment to quality, safety, and innovation supports long-term viability and competitiveness in Malaysia's growing consumer electronics and smart-device market. With effective planning and continued investment in R&D, ThermaCorex has the potential to expand regionally and evolve into a leading player in smart thermal technology.

## 13.0 Recommendations

To strengthen financial performance and accelerate the company to a path for long-term sustainability, ThermaCorex can consider the following strategies:

### 1. Strengthening Cost Management

- Optimise production processes to reduce unit manufacturing cost.
- Source heating elements, batteries, and sensors from high-volume suppliers to lower component expenses.
- Implement quality control systems to minimise returns and warranty costs.

### 2. Expanding Revenue Streams

- Introduce complementary products such as smart travel charging kits, and temperature-adaptive picnic bag.
- Provide corporate gift packages for companies, universities, and events.
- Explore licensing the heating control algorithm to industries such as food delivery, outdoor equipment, or medical warming devices.

### 3. Focusing on Financial Planning

- Build a financial buffer to manage unexpected cost spikes in manufacturing or logistics.
- Reinvest early profits into R&D for improved heating efficiency, battery capacity, and safety features.

- Adopt structured budgeting to support long-term growth and product line expansion.

#### **4. Leveraging Partnerships**

- Collaborate with government agencies supporting hardware innovation and green technology to secure grants or incentives.
- Partner with universities and research institutions for testing new materials and energy-efficient designs.
- Engage local distributors and retailers to enhance product availability nationwide.

#### **5. Preparing for Market Expansion**

- Identify new regional markets in Southeast Asia with demand for smart, portable lifestyle devices.
- Adapt heating settings and design variations for different climate conditions or user groups.
- Explore export opportunities to outdoor retail chains and premium lifestyle product distributors.

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